Hydrohalogenation Reaction of Substituted 1,2-Allenic Carboxylic Acids, Esters, Amides, Nitriles, and Diphenyl Phosphine Oxides

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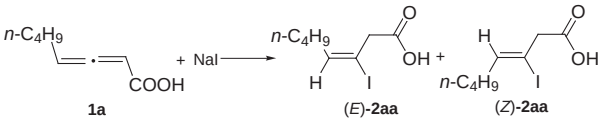
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Abstract: The hydrohalogenation reaction of 1,2-allenic carboxylic acids, esters, amides, nitriles, and diphenyl phosphine oxides with HX or MX (X = I, Br, Cl, M = Na, Li) in different solvents, such as HOAc, CF_3CO_2H, and CF_3CO_2H-HOAc (1:1) was studied. The reaction is a nucleophilic conjugate addition in nature and provides efficient entries to 3-halo-3-enoic acids, esters, amides, nitriles, and 2-haloallyl diphenyl phosphine oxides. The β,γ- over α,β-selectivity and stereoselectivity depend on the solvent(s) and the temperature applied. In most cases, the *Z*- and *E*-isomers can be easily separated by chromatography on silica gel. The subsequent cross-coupling reaction with organometallic reagents and terminal alkynes showed the synthetic potential of these products.

Key words: electron-deficient allenes, hydrohalogenation, regioselectivity, stereoselectivity, solvent effect

3-Alkenoic acids as well as their derivatives and 2-haloallyl phosphine oxides are classes of compounds with synthetic importance due to the uniquely located carbon-carbon double bonds. Recently during the course of our study on the chemistry of allene,[1,2] we developed the hydrohalogenation reaction of electron-deficient allenes,[3] such as 1,2-allenyl ketones,[4,5] carboxylic acids,[4] esters,[4] sulfoxides,[6] and sulfones[7] to afford β-halo-β,γ-unsaturated functionalized alkenes. A challenge encountered here is the potential migration of the C=C bond from the β-halo-β,γ-position to the α,β-position, thus, affording a synthetically useless mixture of β,γ-unsaturated products and α,β-unsaturated products.[4,5,8] In a preliminary communication[9] we have demonstrated that the hydrohalogenation reaction of 2-alkyl-2,3-dienoates with MX in CF_3COOH-HOAc or CF_3COOH afforded 2-alkyl-3-halo-3-alkenoates with a high selectivity. Here, the reaction medium is the key to avoid the formation of a mixture of β,γ-unsaturated and α,β-unsaturated products. In this paper we wish to report our recent results of the hydrohalogenation reaction of differently substituted 2,3-dienoic acids, esters, amides, nitriles, and 1,2-allenyl phosphine oxides and the corresponding synthetic application of the β,γ-unsaturated 3-halo-3-alkenoates, β,γ-unsaturated 3-halo-2-alkenenitriles, and β,γ-unsaturated 2-halo-2-propenyl diphenyl oxides.

Hydrohalogenation Reaction of Substituted 1,2-Dienyl Carboxylic Acids. The results of the reaction of 2,3-octadienoic acid with NaI under different conditions are summarized in Scheme 1.



1a (*E*)-**2aa** (*Z*)-**2aa**

Scheme 1

NaI (equiv)	Solvent	Time (h)	Temp (°C)	Yield (%)[a]	(*Z*/*E*)[a]
1.2	HOAc	40	r.t.	87	4.3:1
2	THF	24	50	94[b]	5.7:1
2	THF	4.5	reflux	94[b]	8.5:1
1.2	HOAc	6.5	reflux	95	9.4:1

[a] Yield and ratio of *Z* and *E* were determined by ^{1}H NMR analysis using CH_2Br_2 as the internal standard. [b]10 equiv of HOAc were added.

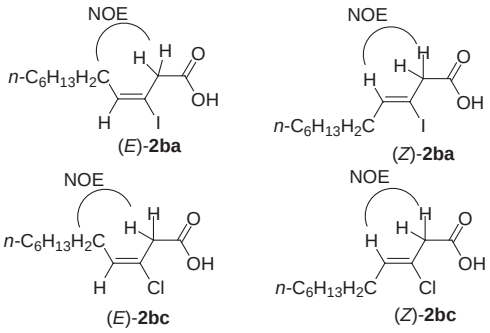


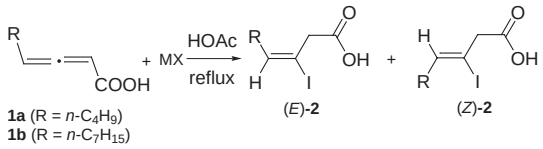
Figure 1

Scheme 1 shows that the reaction already occurs at room temperature but the stereoselectivity is higher at a higher reaction temperature. After 6.5 h under reflux the reaction afforded product **2** in 95 % yield with a *Z*/*E* ratio of 9.4:1; the products (*E*)-**2aa** and (*Z*)-**2aa** can be separated by column chromatography on silica gel, providing synthetically useful entries to both *Z*- and *E*-isomers of 3-iodo-2-alkenoic acids and derivatives.

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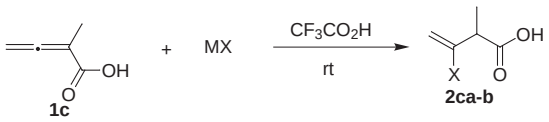
Table 1 Hydrohalogenation Reaction of 2,3-Alkadienoic Acids with MX



1a (R = *n*-C_4H_9)
1b (R = *n*-C_7H_{15})

Entry	1	MX	Time (h)	Product	Yield (%)	(*Z*/*E*)
1	**1a**	LiBr·H_2O	6.5	**2ab**	89	70:30
2	**1a**	LiCl·H_2O	6.5	**2ac**	87	37.5:62.5
3	**1b**	NaI	7	**2ba**	99	80:20
4	**1b**	LiBr·H_2O	7	**2bb**	82	64:36
5	**1b**	LiCl·H_2O	7	**2bc**	87	43:57

The corresponding reaction of 2,3-octadienoic acid or 2,3-nonadienoic acid with LiBr⟨H_2O or LiCl·H_2O afforded a lower ratio of two stereoisomers (*Z*)-**2** and (*E*)-**2** (entries 1, 2, 4, and 5, Table 1). The two stereoisomers of the corresponding bromides can also be separated by column chromatography on silica gel. The stereochemistry was determined by the 2D NOESY study of (*E*)-**2ba**, (*Z*)-**2ba,** and (*E*)-**2bc** (Figure 1).



MX	Time (h)	Yield (%)
NaI	2.5	72 (**2ca**)
LiBr·H_2O	3	61 (**2cb**)

Scheme 2

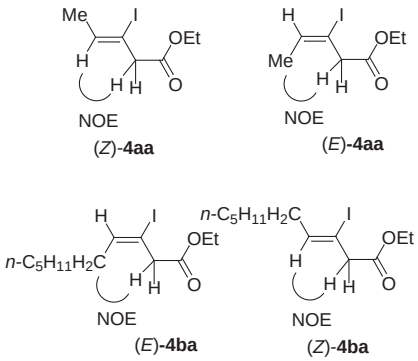


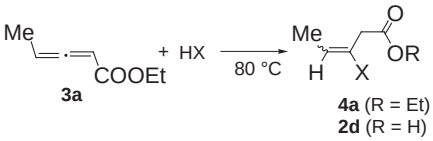
Figure 2

The hydrohalogenation reaction of 2-methyl-substituted 2,3-propadienoic acid **1c** with MX (M = Na, Li) in CF_3CO_2H afforded the corresponding products in good yields (Scheme 2).

Hydrohalogenation Reaction of Substituted Dienoates with MX. The reaction of ethyl 2,3-pentadienoates with NaI, LiBr·H_2O, and LiCl·H_2O in HOAc went smoothly and afforded the corresponding ethyl 3-halo-3-pentenoates in high yields, although the stereoselectivity is not high (Table 2).

The configuration of the carbon-carbon double bond in 3-halo-3-pentenoates **4aa** and **4ba** was determined by 2D NOESY experiments (Figure 2).

However, a major side reaction, i.e., the hydrolysis of the ester group to carboxylic acid, was observed when ethyl-2,3-pentadienoate was treated with HX.[10] It afforded a mixture of ethyl 3-halo-3-pentenoate and 3-halo-3-pentenoic acid (Scheme 3).



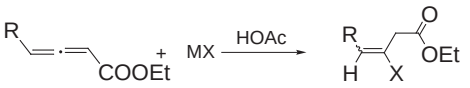
Scheme 3

X	Time (h)	Yield of **4a** (%)	(*Z*/*E*)[a]	Yield of **2d** (%)	(*Z*/*E*)[a]
I	4	40 (**4aa**)	2.3:1	48 (**2da**)	1.05:1
Br	24	trace (**4ab**)		58 (**2db**)	1.7:1
CI	48	trace (**4ac**)		83 (**2dc**)	1:1.1

[a] Z/*E* ratio determined by ^{1}H NMR analysis

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Table 2 Hydrohalogenation Reaction of 2,3-Dienoates **3a** and **3b**



3a (R = Me)
3b (R = *n*-C_6H_{13})

4a
4b

Entry	**3**	MX (equiv)	Solvent	Temp (°C)	Time (h)	*Z*/*E* ratio	Yield (%)
1	**3a**	NaI (2)	HOAc	80	6	4.7:1	94 (**4aa**)
2	**3a**	NaI (1.5)	CF_3CO_2H	r.t.	6	1:2.5	76 (**4aa**)
3	**3a**	LiBr·H_2O (2)	HOAc	80	24	5.5:1	83 (**4ab**)
4	**3a**	LiCl·H_2O (2)	HOAc	80	38	1:1.3	70 (**4ac**)
5	**3b**	NaI (1.5)	CF_3CO_2H	r.t.	1	1:3	98 (**4ba**)
6	**3b**	LiBr·H_2O (1.5)	CF_3CO_2H	r.t.	2.5	1:3	90 (**4bb**)
7	**3b**	LiBr·H_2O (1.5)	HOAc	80	29	1:1	72 (**4bb**)
8	**3b**	LiCl·H_2O (3)	CF_3CO_2H	r.t.	54	-	0[a]
9	**3b**	LiCl·H_2O (3)	CF_3CO_2H	reflux	2.1	1:3	84 (**4bc**)

[a] No reaction.

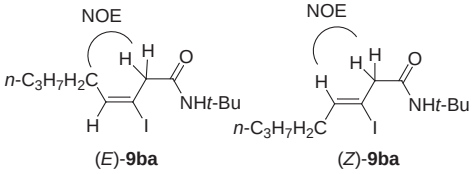
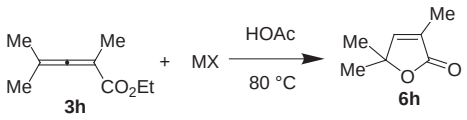


Figure 3



Scheme 4

MX (equiv)	Solvent	Time	**6h**,Yield (%)
LiBr (2)	HOAc	17 h	65
LiBr (2)	HOAc-CF_3CO_2H	10 min	67
LiBr (2)	CF_3CO_2H	10 min	57

The slow hydrohalogenation reaction of ethyl 2-methyl-2,3-butadienoate with MX in HOAc afforded slowly a mixture of 2-enoates and 3-enoates in low yields and selectivity (Table 3). However, when a 1:1 mixture of CF_3CO_2H and HOAc or CF_3CO_2H alone was used as the reaction medium, the hydrohalogenation went smoothly to afford ethyl 3-halo-2-methyl-3-butenoates in fairly good yields and excellent selectivity (compare entries 1–5 with 6–12, Table 3), while the corresponding reaction with HX was reported to afford a mixture of β,γ-unsaturated alkenoate and α,β-unsaturated alkenoate.[11] The reaction in CF_3CO_2H was much more efficient that in CF_3CO_2H-HOAc (1:1) (compare entries 8 with 6 and 9 with 7, Table 3).

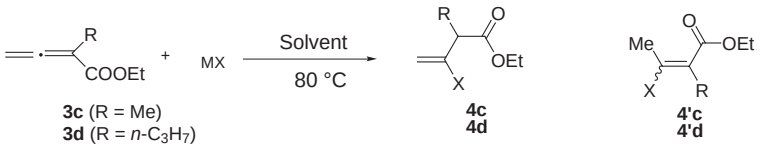
For trisubstituted allenes, i.e., 2,3-dienoates **3e-g**, the reaction also went smoothly to afford the corresponding β,γ-unsaturated 3-enoates **4** (Table 4). However, with 2,4-disubstituted 2,3-dienoate **3e** and 4,4-disubstituted 2,3-dienoate **3g**, self-cyclization products, i.e., butenolides **6e** and **6g,**[12] were formed as the by-products along with the expected hydrohalogenation products **4ec**, **4ga** and **4gb**. When fully substituted 2,3-dienoate **3h** was treated with MX in HOAc, self-cyclization affording butenolide **6h**[12] was the only reaction observed (Scheme 4).

3-Halo-3-alkenoates are expected to be very useful synthetic intermediates, due to the existing carbon-halogen bond, ester group, carbon-carbon double bond, and the acidic α-proton. The carbon-iodine bond in ethyl 2-methyl-3-iodo-3-butenoate (**4ca**) was shown to cross-couple with terminal alkynes[13–15] and organozinc reagents[15,16] under catalytic conditions using Pd(0) to afford a series of substituted β,γ-unsaturated alkenoates (Scheme 5). Previously, **7b**-type products were prepared via either the alkylation reaction of β-aryl-β-methyl acrylates[17] or dehydration reaction of β-hydroxycarboxylic acids which could be synthesized with a Reformatsky reaction,[18] while the syntheses of **7a**- and **7c**-type compounds have not been reported so far.

Hydrohalogenation Reaction of 2,3-Dienamides. The results of hydrohalogenation reactions of 2,3-dienamides **8a**–**8d** are summarized in Table 5. The stereoselectivity of the hydrohalogenation reaction was determined by the NOESY experiments of (*E*)-**9ba** and (*Z*)-**9ba**.

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Table 3 Hydrohalogenation Reaction of 2-Substituted 2,3-Allenoates



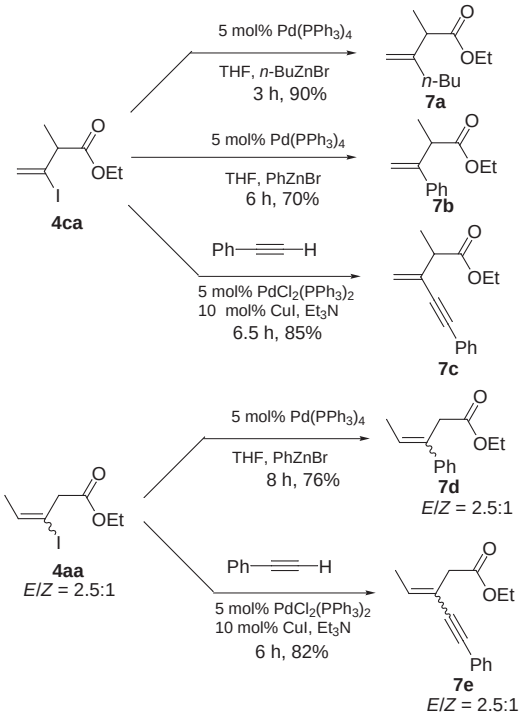
3c (R = Me)
3d (R = *n*-C_3H_7)

4c
4d

4'c
4'd

Entry	MX (equiv)	3	Solvent	Time (h)	4 (%)[a]	4' (%)[a]
1	NaI (1.5)	**3c**	HOAc	24	**4ca** (41)	**4'ca** (10)
2	NaI (4)	**3c**	HOAc	72	**4ca** (53)[b]	**4'ca** (14)[b]
3	LiBr·H_2O (1.3)	**3c**	HOAc	72	**4cb** (<5)	**4'cb** (<5)
4	LiBr·H_2O (1.5)[c]	**3c**	HOAc	120	**4cb** (<5)	**4'cb** (<5)
5	LiCl·H_2O (1.8)	**3c**	HOAc	74	**4cc** (˜10)	**4'cc** (˜5)
6	NaI (3)	**3c**	HOAc-CF_3CO_2H	4	**4ca** (95)	[d]
7	LiBr·H_2O (3)	**3c**	HOAc-CF_3CO_2H	12	**4cb** (80)	[d]
8	NaI (2)	**3c**	CF_3CO_2H	10 min	**4ca** (90)	[d]
9	LiBr·H_2O (2)	**3c**	CF_3CO_2H	40 min	**4cb** (88)	[d]
10	NaI (2)	**3d**	CF_3CO_2H	20 min	**4da** (98)	[d]
11	LiBr·H_2O (2)	**3d**	CF_3CO_2H	50 min	**4db** (76)	[d]
12	LiCl·H_2O (3)	**3d**	CF_3CO_2H	10	**4dc** (63)	[d]

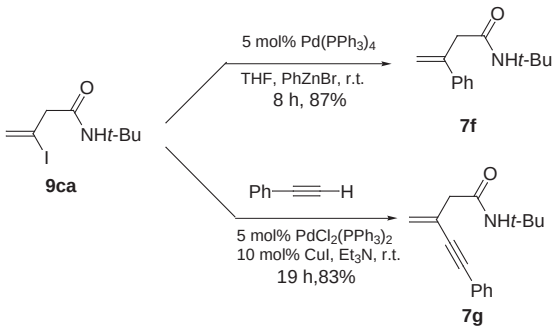
[a] The numbers in parentheses are isolated yields. [b] NMR yield. [c] The reaction was carried out at 40 °C. [d] Not observed.



Scheme 5

From Table 5, it can be concluded that the hydrohalogenation of 2,3-dienamides usually affords the corresponding 3-halo-3-alkenamides in high yields. Both MX-HOAc and aqueous HX systems can be applied. Even for *N*-benzyl-2-methyl-2,3-dienamide (**8d**), the migration of β,g-carbon-carbon double bond to a,b-carbon-carbon double bond was not observed (entries 13–18, Table 5). When 2,3-dienamides were treated with MX in CF_3CO_2H, the reaction can occur even at r.t. (entries 7, 13 and 14, Table 5). The *E*- and *Z*-isomers are readily separated by chromatography on silica gel.

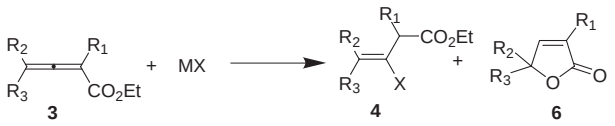
The corresponding Sonogashira coupling and Negishi coupling reactions of **9ca** will provide different β-substituted 3-alkenamide (Scheme 6).



Scheme 6

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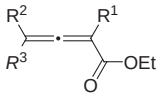
Table 4 Hydrohalogenation Reaction of 2,3-Dienoates



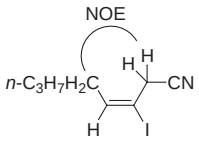
3 **4** **6**

Entry	**3**	MX (equiv)	Solvent	Temp. (°C)	Time (h)	**4** (%)[a]	(*E*/*Z*)
1	**3e**	NaI(1.5)	CF_3CO_2H	r.t.	1.1	**4ea** (78)	3:2
2	**3e**	LiBr·H_2O (1.5)	CF_3CO_2H	r.t.	3	**4eb** (89)	3:2
3	**3e**	LiCl·H_2O (3)	CF_3CO_2H	r.t.	54	0	
4	**3e**	LiCl·H_2O (3)	CF_3CO_2H	reflux	17	**4ec** (63)[b]	3:2
5	**3f**	NaI (1.5)	CF_3CO_2H	r.t.	50 min	**4fa** (78)	1:1
6	**3f**	NaI (2)	CF_3CO_2H	reflux	10 min	**4fa** (58)	1:1.2
7	**3f**	LiBr·H_2O (1.5)	CF_3CO_2H	r.t.	2.5	**4fb** (74)	(1:1)
8	**3f**	LiBr·H_2O (2)	CF_3CO_2H	reflux	20 min	**4fb** (71)	1:1.3
9	**3f**	LiBr·H_2O (2)	CF_3CO_2H-HOAc	80	20 min	**4fb** (63)	1:2.2
10	**3f**	LiCl·H_2O (4)	CF_3CO_2H	reflux	40 min	**4fc** (55)	1.5:1
11	**3f**	LiCl·H_2O (3)	CF_3CO_2H	r.t.	2	**4fc** (39)	2:1
12	**3g**	NaI(2)	CH_3CO_2H	80	4	**4ga** (23)[c]	
13	**3g**	LiBr·H_2O (2)	CH_3CO_2H	80	15	**4gb** (23)[d]	

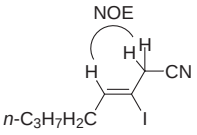
[a] Isolated yield. [b] 19 % of **6e** (R^2 = H, R^3 = *n*-C_6H_{13}, R^1 = Me) was isolated. [c] 15 % of **6g** (R^2 = R^3 = Me, R^1 = H) was isolated. [d] 42 % of **6g** was isolated.



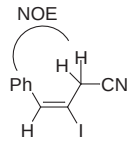
R^1 = Me, R^2 = H, R^3 = *n*-C_6H_{13} **3e**
R^1 = Me, R^2 = H, R^3 = Me **3f**
R^1 = H, R^2 = R^3 = Me **3g**



(*E*)-**11ca**



(*Z*)-**11ca**



(*E*)-**11da**

Figure 4

Hydrohalogenation Reaction of 2,3-Dienenitriles. The results of the corresponding 2,3-dienenitriles were summarized in Table 6.

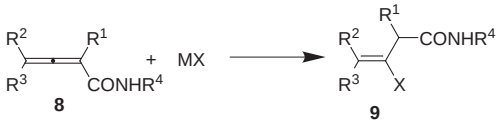
For the reactions of 2,3-dienenitriles, the hydrochlorination and hydrobromination with LiX in HOAc were low-yielding, however, the yields were improved using a 1:1 mixture of CF_3CO_2H-HOAc (compare entries 3–5 with 6–8, Table 6). With 4-phenyl- 2,3-butadienenitrile (**10d**), only the *E*-isomers were formed, probably due to the existence of the phenyl group. The stereoselectivity of the hydrohalogenation reaction was determined by the NOESY study of *(E)*-**11ca,** (*Z*)-**11ca** and (*E*)-**11da** (Figure 4).

Isomerization of (*E*)-**11ca** and (*Z*)-**11ca** in HOAc in the presence of NaI at 70 °C was not observed, indicating the direct dependence of the stereoselectivity on the solvent used (compare entries 12 and 13, Table 6).

The corresponding Sonogashira coupling and Negishi coupling reactions will provide different β-substituted 3-alkenenitrile (Scheme 7).

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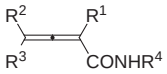
Table 5 Hydrohalogenation Reaction of 2,3-Dienamides



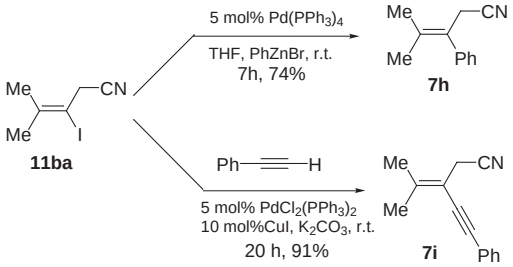
8 9

Entry	8	MX	Solvent	Temp (°C)	Time (h)	Yield (%)
1	**8a**	NaI	HOAc	70	4	**9aa** (87)
2	**8a**	LiBr·H_2O	HOAc	70	4	**9ab** (96)
3	**8a**	LiCl·H_2O	HOAc	70	7	**9ac** (75)
4	**8b**	NaI	HOAc	r.t.	4	**9ba** [*E* (36), *Z* (62)]
5	**8b**	LiBr·H_2O	HOAc	70	22	**9bb** [*E* (34), *Z* (56)]
6	**8b**	LiCl·H_2O	HOAc	70	22	**9bc** [*E* (56), *Z*(23)]
7	**8b**	NaI	CF_3CO_2H	r.t.	0.5	**9ba** [*E* (64), *Z* (34)]
8	**8b**	LiBr·H_2O	CF_3CO_2H	70	4	**9bb** [*E* (48), *Z* (21)]
9	**8b**	LiCl·H_2O	CF_3CO_2H	70	26	**9bc** [*E* (15), *Z* (8)]
10	**8c**	NaI	HOAc	70	18	**9ca** (97)
11	**8c**	LiBr·H_2O	HOAc	70	23	**9cb** (93)
12	**8c**	LiCl·H_2O	HOAc	70	23	**9cc** (91)
13	**8d**	NaI	CF_3CO_2H	r.t.	1	**9da** (97)
14	**8d**	LiBr·H_2O	CF_3CO_2H	r.t.	24	**9db** (99)
15	**8d**	LiCl·H_2O	CF_3CO_2H	40	70	**9dc** (87)
16	**8d**	aq HI	aq HI	r.t.	11	**9da** (77)
17	**8d**	aq HBr	aq HBr	r.t.	11	**9db** (86)
18	**8d**	aq HCl	aq HCl	r.t.	11	**9dc** (94)

The structures of these compounds should be shown as below.



R^1 = H, R^2 = R^3 = Me, R^4 = H **8a**
R^1 = H, R^2 = *n*-Bu, R^3 = H, R^4 = *t*-Bu **8b**
R^1 = R^2 = R^3 = H, R^4 = *t*-Bu **8c**
R^1 = Me, R^2 = R^3 = H, R^4 = Bn **8d**

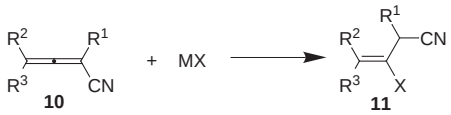


Scheme 7

Hydrohalogenation Reaction of 1,2-Allenyl Diphenyl Phosphine Oxides. A series of 1,2-allenyl diphenyl phosphine oxides was synthesized.[19] In case of **12b** and **12c** (Scheme 8) the reactivities towards MX in HOAc were low. The reaction of propadienyl diphenyl phosphine oxide **12a** with MX (M = Na, Li, X = Cl, Br, I) afforded 2-halo-2-propenyl diphenyl phosphine oxides **13a-c** in reasonable yields, while the similar reaction with aqueous HX afforded the products in low yields (compare entries 5–7 with 1–4, Table 7).

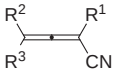
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Table 6 Hydrohalogenation Reaction of 1,2-Allenyl Cyanides



Entry	10	MX	Solvent	Temp (°C)	Time (h)	Yield (%)
1	**10a**	LiBr·H_2O	HOAc	70	21	**11ab** (86)
2	**10a**	LiCl·H_2O	HOAc	70	24	**11ac** (78)
3	**10b**	NaI	CF_3CO_2H-HOAc	70	1.5	**11ba** (80)
4	**10b**	LiBr·H_2O	CF_3CO_2H-HOAc	70	7.5	**11bb** (78)
5	**10b**	LiCl·H_2O	CF_3CO_2H-HOAc	70	8.5	**11bc** (76)
6	**10b**	NaI	HOAc	70	12	**11ba** (37)
7	**10b**	LiBr·H_2O	HOAc	85	7.5	**11bb** (17)
8	**10b**	LiCl·H_2O	HOAc	80	21	**11bc** (18)
9	**10c**	NaI	CF_3CO_2H-HOAc	70	3.5	**11ca** [*E* (63), *Z* (16)]
10	**10c**	LiBr·H_2O	CF_3CO_2H-HOAc	70	6	**11cb** [*E* (58), *Z* (22)]
11	**10c**	LiCl·H_2O	CF_3CO_2H-HOAc	70	8	**11cc** [*E* (50), *Z* (22)]
12	**10c**	NaI	HOAc	70	21	**11ca** [70 (*E*:*Z* = 1:1)]
13	**10c**	NaI	CF_3CO_2H	70	10 min	**11ca** [58 (*E*:*Z* = 3.1:1)]
14	**10d**	NaI	CF_3CO_2H-HOAc	70	5	(*E*)-**11da** (44)
15	**10d**	LiBr·H_2O	CF_3CO_2H-HOAc	70	5	(*E*)-**11db** (82)
16	**10d**	LiCl·H_2O	CF_3CO_2H-HOAc	70	6	(*E*)-**11dc** (37)

The structures of these compounds should be shown as below.



$R^1 = R^2 = R^3 = H$ **10a**
$R^1 = H$, $R^2 = R^3 = Me$ **10b**
$R^1 = H$, $R^2 = n$-Bu, $R^3 = H$ **10c**
$R^1 = H$, $R^2 = Ph$, $R^3 = H$ **10d**

In conclusion, we have developed a convenient method for the synthesis of 3-halo-3-alkenoic acids, their derivatives and 2-halo-2-propenyl diphenyl phosphine oxides via a simple reaction of 2,3-dienoic acids, their derivatives, and 1,2-allenyl diphenyl phosphine oxides with MX in HOAc, CF_3CO_2H-HOAc, or CF_3CO_2H. In most cases, the yields are high; the stereoselectivity depends on the reaction temperature and the solvent used; the ratios of β,γ-unsaturated and α,β-unsaturated products are determined by the acidity of solvent used. The uniquely located C=C bonds, C-X bonds, and the electron-withdrawing functional groups make these products synthetically useful. In most cases, the *Z*- and *E*-isomers can be readily separated providing efficient entries to both relevent stereoisomers. The reaction is nucleophilic in nature.[4,5]

Synthesis of Starting Materials

1,2-Allenyl carboxylic acids **1a,**[19] **1b,**[19] **1c,**[20,3e] 1,2-allenyl carboxylates **3a-h,**[20] 2,3-dienamides **8a**,[21] **8b**,[21] **8c**,[21] **8d**,[22] 2,3-dienitriles **10a**,[19,23] **10b**, [19,23] **10c**,[19,23] **10d**,[19,23] and 1,2-dienyl diphenyl phosphine oxides **12a**–**12c**[19] were synthesized according to the reported procedures.

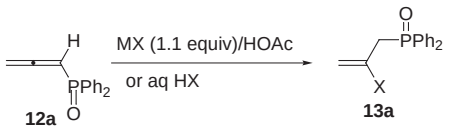
Hydrohalogenation Reaction of 2,3-Dienoic Acids

Iodo-3-octenoic acid (2aa); **Typical Procedure**

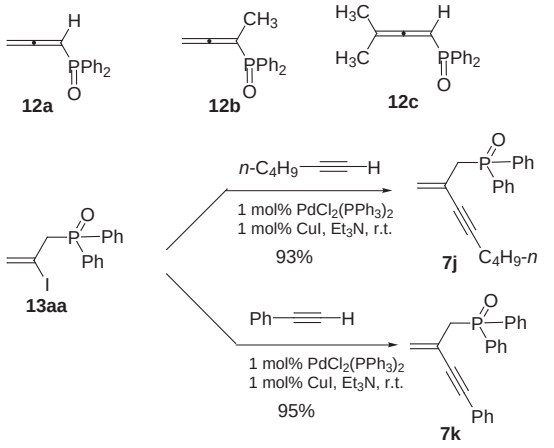
To a solution of 2,3-octadienoic acid (105 mg, 0.75 mmol) in 1.5 mL of glacial acetic acid was added NaI (135 mg, 0.90 mmol) under Ar. The reaction mixture was heated and refluxed for 6.5 h upon completion of the reaction as monitored by TLC. HCl (2 N, 0.4 mL) was added (for acids only!), and the mixture was quenched with H_2O (1 mL) and then extracted with Et_2O (5 × 5 mL). The combined organic layer was washed subsequently with aq $Na_2S_2O_3$ (5%) (for

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Table 7 Hydrohalogenation Reaction of 1,2-Propadienyl Diphenyl Phosphine Oxide



Entry	MX	Solvent	Temp (°C)	Time (h)	Yield (%)
1	NaI	HOAc	reflux	23	**13aa** (60)
2	LiI	HOAc	100	12	**13aa** (75)
3	LiBr·HO	HOAc	100	36	**13ab** (70)
4	$LiCl·H_2O$	HOAc	100	40	**13ac** (58)
5	aq HI	aq HI	reflux	48	**13aa** (46)
6	aq HBr	aq HBr	reflux	60	**13ab** (37)
7	aq HCl	aq HCl	reflux	60	**13ac** (31)



Scheme 8

iodides only!), sat. NH_4Cl and dried over $MgSO_4$. The solvent was removed and the crude product was purified by flash chromatography (eluent: petroleum ether-ethyl acetate, 5:1) to afford 135 mg (67%) of pure (*Z*)-**2aa** and 57 mg (28%) of the *Z*- and *E*-mixture **2aa** (*Z*/*E* = 67.5:32.5).

(*Z*)-**2aa**: yellow oil.

IR (neat): ν = 3200 (bs), 2960, 1700, 1650, 940 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.70 (t, *J* = 6.6 Hz, 1 H), 3.67 (s, 2 H), 2.14 (q, *J* = 7.2 Hz, 2 H), 1.5~1.2 (m, 4 H), 0.90 (t, *J* = 6.6 Hz, 3 H), the following data are discernible for the *E*-isomer: δ = 6.42 (t, *J* = 7.5 Hz, 1 H), 3.65 (s, 2 H), 2.05 (q, *J* = 7.0 Hz, 2 H), 1.5~1.2 (m, 4 H), 0.90 (t, *J* = 6.6 Hz, 3 H).

MS *m/z* (%): 267 (4, $M^+ - 1$), 81 (100).

HRMS: calcd for $C_8H_{13}IO_2$,: 267.9959. Found: 267.9946.

3-Bromo-3-octenoic Acid (2ab)

Starting from **1a** (99 mg, 0.71 mmol) and $LiBr·H_2O$ (89 mg, 0.85 mmol) in HOAc (0.5 mL), the mixture was heated under reflux for 6.5 h to afford 139 mg (89%) of **2ab** (*Z*/*E* = 70:30).

(*Z*)-**2ab**: liquid.

IR (neat): ν = 3200 (bs), 2970, 1700, 1650 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 9.10 (br s, 1H), 5.85 (t, *J* = 6.8 Hz, 1 H), 3.52 (s, 2 H), 2.19 (q, *J* = 7.2 Hz, 2 H), 1.5~1.2 (m, 4 H), 0.90 (t, *J* = 7.4 Hz, 3 H), the following data are discernible for the (*E*)-isomer: δ = 9.10 (bs, 1H), 6.11 (t, *J* = 7.7 Hz, 1 H), 3.57 (s, 2 H), 2.04 (q, *J* = 7.0 Hz, 2 H), 1.5~1.2 (m, 4 H), 0.91 (t, *J* = 7.5 Hz, 3 H).

MS *m/z* (%): 223 [1, $M^+(^{81}Br) + 1$], 221 [1, $M^+(^{79}Br) + 1$], 81 (100).

HRMS: calcd for $C_8H_{13}{}^{79}BrO_2$: 220.0099. Found: 220.0134, HRMS: calcd for $C_8H_{11}{}^{81}BrO$ [$M^+(^{81}Br) - H_2O$]: 203.9973. Found: 203.9975, HRMS: calcd. for $C_8H_{11}{}^{79}BrO$ [$M^+(^{79}Br) - H_2O$]: 201.9993. Found: 202.0011.

3-Chloro-3-octenoic Acid (2ac)

Starting from **1a** (107 mg, 0.76 mmol) and $LiCl·H_2O$ (55.4 mg, 0.92 mmol) in HOAc (0.5 mL), the mixture was heated under reflux for 6.5 h to afford 117 mg (87%) of **2ac** (*Z*/*E* = 37.5:62.5).

(*Z*)-**2ac**: liquid.

IR (neat): ν = 3200 (bs), 2970, 1710, 1650 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 8.0 (br s, 1 H), 5.66 (t, *J* = 7.8 Hz, 1 H), 3.36 (s, 2 H), 2.14 (q, *J* = 7.0 Hz, 2 H), 1.5~1.2 (m, 4 H), 0.88 (t, *J* = 7.0 Hz, 3 H), the following data are discernible for the (*E*)-isomer: δ = 8.0 (bs, 1 H), 5.85 (t, *J* = 7.0 Hz, 1 H), 3.43 (s, 2 H), 1.99 (q, *J* = 7.3 Hz, 2 H), 1.5~1.2 (m, 4 H), 0.88 (t, J = 6.7 Hz, 3 H).

MS *m/z* (%): 179 [0.4, $M^+(^{37}Cl) + 1$], 177 [1.0, $M^+(^{35}Cl) + 1$], 141 (100).

HRMS: calcd for $C_8H_{13}{}^{37}ClO_2$: 178.0575. Found: 178.0536, HRMS: calcd for $C_8H_{13}{}^{35}ClO_2$: 176.0604. Found: 176.0628.

3-Iodo-3-undecenoic Acid (2ba)

Starting from **1b** (91 mg, 0.50 mmol) and NaI (90 mg, 0.60 mmol) in HOAc (0.4 mL), the mixture was heated under reflux for 7 h to afford 153 mg (9%) of **2ba** (*Z*/*E* = 80:20).

(*Z*)-**2ba**: liquid.

IR (neat): ν = 3200 (bs), 2500, 1700, 1670, 940 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.63 (t, *J* = 6.6 Hz, 1 H), 3.61 (s, 2 H), 2.07 (q, *J* = 7.1 Hz, 2 H), 1.50–1.22 (m, 2H), 1.22~1.10 (m, 8 H), 0.82 (t, *J* = 6.2 Hz, 3 H).

MS *m/z* (%): 311 (7, $M^+ + 1$), 123 (100).

HRMS: calcd for $C_{11}H_{19}IO_2$: 310.0429. Found: 310.0424.

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(*E*)-**2ba**: liquid.

IR (neat): ν = 3200 (bs), 2690, 1700 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.37 (t, *J* = 7.6 Hz, 1 H), 3.60 (s, 2 H), 1.99 (q, *J* = 7.3 Hz, 2 H), 1.50–1.22 (m, 2H), 1.22~1.10 (m, 8 H), 0.78 (t, *J* = 6.5 Hz, 3 H).

MS *m/z* (%): 309 (2, $M^+ - 1$), 123 (100).

HRMS: calcd for $C_{11}H_{19}IO_2$: 310.0429. Found: 310.0425.

3-Bromo-3-undecenoic Acid (2bb)
Starting from **1b** (91 mg, 0.50 mmol) and $LiBr{\cdot}H_2O$ (63 mg, 0.60 mmol) in HOAc (0.4 mL), the mixture was heated under reflux for 7 h to afford 108 mg (82%) of **2bb** (*Z*/*E* = 64:36).

(*Z*)-**2bb**: liquid.

IR (neat): ν = 3200 (bs), 2950, 1700, 1670 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 10.00 (br s, 1H), 5.79 (t, *J* = 6.8 Hz, 1 H), 3.51 (s, 2 H), 2.12 (q, *J* = 7.0 Hz, 2 H), 1.50–1.32 (m, 3 H), 1.32~1.1 (m, 8 H), 0.81 (t, *J* = 6.5 Hz, 3 H).

MS *m/z* (%): 265 [2, $M^+(^{81}Br)$ + 1], 263 [2, $M^+(^{79}Br)$ + 1], 123 (100).

HRMS: calcd for $C_{11}H_{19}{}^{81}BrO_2$: 264.0548. Found: 264.0517, HRMS: calcd for $C_{11}H_{19}{}^{79}BrO_2$: 262.0569. Found: 262.0592.

(*E*)-**2bb**: liquid.

IR (neat): ν = 3200 (bs), 2900, 1700, 1670 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 10.00 (br s, 1H), 6.05 (t, *J* = 7.7 Hz, 1 H), 3.46 (s, 2 H), 1.98 (q, *J* = 7.3 Hz, 2 H), 1.50–1.32 (m, 2H), 1.32~1.10 (m, 8 H), 0.81 (t, *J* = 6.5 Hz, 3 H).

MS *m/z* (%): 265 [9, $M^+(^{81}Br)$ + 1], 263 [9, $M^+(^{79}Br)$ + 1], 123 (100).

HRMS: calcd for $C_{11}H_{19}{}^{81}BrO_2$: 264.0548. Found: 264.0570, HRMS: calcd for $C_{11}H_{19}{}^{79}BrO_2$: 262.0569. Found: 262.0554.

3-Chloro-3-undecenoic Acid (2bc)
Starting from **1b** (146 mg, 0.80 mmol) and $LiCl{\cdot}H_2O$ (58 mg, 0.96 mmol) in HOAc (0.25 mL), the mixture was heated under reflux for 7 h to afford 149 mg (85%) of **2bc** (*Z*/*E* = 43:57).

(*Z*)-**2bc**: liquid.

IR (neat): ν = 3200 (bs), 2900 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.67 (t, *J* = 7.0 Hz, 1 H), 3.37 (s, 2 H), 2.20 (q, *J* = 7.0 Hz, 2 H), 1.5~1.1 (m, 10 H), 0.87 (t, *J* = 6.3 Hz, 3 H), the following data were discernible for *E*-isomer: d = 5.85 (t, *J* = 7.0 Hz, 1 H), 3.43 (s, 2 H), 2.05 (q, *J* = 7.4 Hz, 2 H), 1.5~1.1 (m, 10 H), 0.87 (t, *J* = 6.3 Hz, 3 H).

MS *m/z* (%): 221 [0.2, $M^+(^{37}Cl)$ + 1], 219 [0.5, $M^+(^{35}Cl)$ + 1], 43 (100).

HRMS: calcd for $C_{11}H_{19}{}^{35}ClO_2$: 218.1074. Found: 218.1091, HRMS: calcd for $C_{11}H_{19}O_2$ (M^+-Cl): 183.1385. Found: 183.1359.

2-Methyl-3-iodo-3-butenoic acid (2ca)
Starting from **1c** (20.6 mg, 0.21 mmol) and NaI (63 mg, 0.42 mmol) in CF_3CO_2H (0.5 mL) at r.t. under Ar, 34.1 mg (72%) of **2ca** were afforded after 2.5 h.

2ca: liquid.

IR (neat): ν = 2983, 1703, 1616, 1231 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.30 (s, 1 H), 5.90 (s, 1 H), 3.37 (q, *J* = 6.8 Hz, 1 H), 1.34 (d, *J* = 6.8 Hz, 3 H).

MS *m/z* (%): 226 (57, M^+), 43 (100).

HRMS: calcd for $C_5H_7IO_2$: 225.9485. Found: 225.9473.

2-Methyl-3-bromo-3-butenoic Acid (2cb)
Starting from **1c** (40 mg, 0.41 mmol) and $LiBr{\cdot}H_2O$ (86 mg, 0.82 mmol) in CF_3CO_2H (0.50 mL) at r.t., 44.5 mg (61%) of **2cb** were afforded after 3 h.

2cb: liquid.

IR (neat): ν = 2999, 1712, 1631, 1234 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.85 (bs, 1 H), 5.82 (d, *J* = 1.8 Hz, 1 H), 5.63 (d, *J* = 1.8 Hz, 1 H), 3.54 (q, *J* = 7.0 Hz, 1 H), 1.41 (d, *J* = 7.0 Hz, 3 H).

MS *m/z* (%): 179 [18, $M^+(^{79}Br)$], 181 [17, M^+ (^{81}Br)], 54 (100).

HRMS: calcd for $C_5H_7{}^{81}BrO_2$: 179.9609. Found: 179.9600, HRMS: calcd for $C_5H_7{}^{79}BrO_2$: 177.9624. Found: 177.9618.

Hydrohalogenation Reaction of 1,2-Allenyl Carboxylates

Ethyl 3-Iodo-3-pentenoate (4aa)
Starting from **3a** (72 mg, 0.57 mmol) and NaI (173 mg, 1.15 mmol) in HOAc (0.50 mL), the reaction was carried out at 80 °C for 6 h to afford 136 mg (94%) of **4aa** (*Z*/*E* = 4.7:1).

(*Z*)-**4aa**: liquid.

IR (neat): ν = 3420, 2960, 1725, 1670, 1175, 1020 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.76 (q, *J* = 6.3 Hz, 1 H), 4.20 (q, *J* = 7.2 Hz, 2 H), 3.62 (s, 2 H), 1.79 (d, *J* = 6.2 Hz, 3 H), 1.29 (t, *J* = 7.2 Hz, 3 H), the following data are discernible for the *E*-isomer: δ = 6.47 (q, *J* = 7.1 Hz, 1 H), 4.20 (q, *J* = 7.2 Hz, 2 H), 3.60 (s, 2 H), 1.68 (d, *J* = 7.1 Hz, 3 H), 1.30 (t, *J* = 7.2 Hz, 3 H).

MS *m/z* (%): 254 (15, M^+), 127 (100).

HRMS: calcd for $C_7H_{11}IO_2$: 253.9803. Found: 253.9809.

Starting from **3a** (83 mg, 0.66 mmol) and HI (45%, 0.25 ml) at 80 °C, the reaction afforded 66 mg (40%) of **4aa** (*Z*/*E* = 2.3:1) and 71 mg (48%) of **2da** (*Z*/*E* = 1.05:1) after 4 h.

(*Z*)-**2da**: liquid.

IR: ν = 3200 (bs), 2960, 1705, 1650, 1400, 1030, 1020 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.79 (q, *J* = 7.1 Hz, 1 H), 3.69 (s, 2 H), 1.79 (d, *J* = 7.1 Hz, 3H), the following data are discernible for the *E*-isomer: δ = 6.50 (q, *J* = 6.4 Hz, 1 H), 3.67 (s, 2 H), 1.68 (d, *J* = 6.4 Hz, 3H).

MS *m/z* (%): 226 (M^+), 99 (100).

HRMS: calcd for $C_5H_7IO_2$: 225.9490. Found: 225.9498.

Ethyl 3-Bromo-3-pentenoate (4ab)
Starting from **3a** (58 mg, 0.46 mmol) and $LiBr{\cdot}H_2O$ (100 mg, 0.95 mmol) in HOAc (0.50 mL), the reaction was carried out at 80 °C for 24 h to afford 79 mg (83%) of **4ab** (*Z*/*E* = 5.5:1).

(*Z*)-**4ab**: liquid.

IR (neat): ν = 3400, 2960, 2870, 1740, 1620, 1180, 1040 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.88 (q, *J* = 6.4 Hz, 1 H), 4.20 (q, *J* = 7.1 Hz, 2 H), 3.47 (s, 2 H), 1.78 (d, *J* = 6.2 Hz, 3 H), 1.28 (t, *J* = 7.1 Hz, 3 H), the following data are discernible for the *E*-isomer: δ = 6.10 (q, *J* = 7.1 Hz, 1 H), 4.20 (q, *J* = 7.1 Hz, 2 H), 3.52 (s, 2 H), 1.68 (d, *J* = 7.2 Hz, 3 H), 1.29 (t, *J* = 7.1 Hz, 3 H).

MS *m/z* (%): 209 [43; $M^+(^{81}Br)$ + 1], 207 [43; $M^+(^{79}Br)$ + 1], 53 (100):

HRMS. calcd for $C_7H_{11}{}^{81}BrO_2$: 207.9922. Found: 207.9903, HRMS: calcd for $C_7H_{11}{}^{79}BrO_2$: 205.9942. Found: 205.9918.

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3-Bromo-3-pentenoic Acid (2db)
Starting from **3a** (70 mg, 0.56 mmol) and HBr (40 %, 1 mL), the reaction was carried out at 80 °C for 24 h to afford 58 mg (58%) of **2db** (*Z*/*E* = 1.7:1).

(*Z*)-**2db**: liquid.

IR (neat): ν = 3300 (bs), 1710, 1660 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 8.50 (br s, 1H), 5.95 (q, *J* = 6.4 Hz, 1 H), 3.58 (s, 2 H), 1.78 (d, *J* = 7.1 Hz, 3H), the following data are discernible for the (*E*)-isomer: δ = 8.50 (bs, 1H), 6.18 (q, *J* = 7.1 Hz, 1 H), 3.54 (s, 2 H), 1.68 (d, *J* = 6.5 Hz, 3H).

MS *m/z* (%): 180 [40, $M^+(^{81}Br)$], 178 [40, $M^+(^{79}Br)$], 99 (100).

HRMS: calcd for $C_5H_7{}^{81}BrO_2$: 179.9609. Found: 179.9624, HRMS: calcd for $C_5H_7{}^{79}BrO_2$: 177.9630. Found: 177.9606.

Ethyl 3-Chloro-3-pentenoate (4ac)
Starting from **3a** (83 mg, 0.66 mmol) and $LiCl{\cdot}H_2O$ (80 mg, 1.32 mmol) in HOAc (0.50 mL), the reaction was carried out at 80 °C for 38 h to afford 75 mg (70%) of **4ac** (*E*/*Z* = 1.3:1) as a liquid.

(*Z*)-**4ac**: liquid

IR (neat): ν = 3420, 2940, 1730, 1650, 1175, 1020 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.71 (q, *J* = 6.5 Hz, 1 H), 4.18 (q, *J* = 7.1 Hz, 2 H), 3.31 (s, 2 H), 1.77 (d, *J* = 6.5 Hz, 3 H), 1.27 (t, *J* = 7.1 Hz, 3 H), the following data are discernible for the *E*-isomer: δ = 5.87 (q, *J* = 7.2 Hz, 1 H), 4.19 (q, *J* = 7.1 Hz, 2 H), 3.37 (s, 2 H), 1.69 (d, *J* = 7.1 Hz, 3 H), 1.28 (t, *J* = 7.1 Hz, 3 H).

MS *m/z* (%): 164 [21, $M^+(^{37}Cl)$], 162 [62, $M^+(^{35}Cl)$], 53 (100).

HRMS: calcd for $C_7H_{11}{}^{37}ClO_2$: 164.0418. Found: 164.0390, HRMS: calcd for $C_7H_{11}{}^{35}ClO_2$: 162.0448. Found: 162.0445.

3-Chloro-3-pentenoic Acid (2dc)
Starting from **3a** (70 mg, 0.56 mmol) and HCl (36 %, 1 mL), the reaction was carried outat 80 °C for 48 h to afford 54.2 mg (83%) of **2dc** (*Z*/*E* = 1:1.1).

(*E*)-**2dc**: liquid.

IR (neat): ν = 3300 (bs), 2960, 1710, 1060 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 8.50 (bs, 1H), 5.91 (q, *J* = 7.2 Hz, 1 H), 3.43 (s, 2 H), 1.70 (d, *J* = 7.2 Hz, 3H), the following data are discernible for the *Z*-isomer: δ = 8.50 (bs, 1H), 5.75 (q, *J* = 6.6 Hz, 1 H), 3.38 (s, 2 H), 1.77 (d, *J* = 6.5 Hz, 3H).

MS *m/z* (%): 136 [18, $M^+(^{37}Cl)$], 134 [54, $M^+(^{35}Cl)$], 53 (100).

HRMS: calcd for $C_5H_7{}^{37}ClO_2$: 136.0105. Found: 136.0099, HRMS: calcd for $C_5H_7{}^{35}ClO_2$: 134.0135. Found: 134.0122.

Ethyl 3-Iodo-3-decenoate (4ba)
Starting from **3b** (102 mg, 0.52 mmol) and NaI (113 mg, 0.75 mmol) in CF_3CO_2H (0.50 mL) at r.t.,165 mg (98%) of **4ba** (*E*/*Z* = 3:1) were afforded after 1 h.

(*E*)-**4ba**: liquid.

IR (neat): ν = 2900, 1740, 1360, 1170 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.39 (t, *J* = 7.6 Hz, 1 H), 4.18 (q, *J* = 7.2 Hz, 2 H), 3.58 (s, 2 H), 2.05 (q, *J* = 7.3 Hz, 2 H), 1.47~1.17 (m, 10 H), 0.87 (t, *J* = 6.6 Hz, 3 H).

MS *m/z* (%): 325 (4, $M^+ + 1$), 109 (100).

HRMS: calcd for $C_{12}H_{21}IO_2$: 324.0585. Found: 324.0606.

(*Z*)-**4ba**: liquid.

IR (neat): ν = 2929, 1742, 1368, 1164 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.67 (t, *J* = 6.7 Hz, 1 H), 4.20 (q, *J* = 7.1 Hz, 2 H), 3.61 (s, 2 H), 2.15 (q, *J* = 7.1 Hz, 2 H), 1.5~1.2 (m, 13 H),0.90 (m, 3 H).

MS *m/z* (%): 324 (3, M^+), 109 (100).

HRMS: calcd for $C_{12}H_{21}IO_2$: 324.0585. Found: 324.0552.

Ethyl 3-Bromo-3-decenoate (4bb)
Starting from **3b** (135 mg, 0.69 mmol) and $LiBr{\cdot}H_2O$ (101 mg, 0.97 mmol) in CF_3CO_2H (0.50 mL) at r.t., 172 mg (90 %) of **4bb** (*E*/*Z* = 3:1) were afforded after 2.5 h.

(*E*)-**4bb**: liquid.

IR (neat): ν = 2900, 1740, 1360, 1170, 1030 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.07 (t, *J* = 7.3 Hz, 1 H), 4.18 (q, *J* = 7.1 Hz, 2 H), 3.49 (s, 2 H), 2.03 (q, *J* = 7.3Hz, 2 H), 1.45~1.15 (m, 11 H), 0.87 (t, *J* = 6.7 Hz, 3 H), the following data are discernible for *Z*-isomer: δ = 5.83 (t, *J* = 7.2 Hz, 1H), 4.18 (q, *J* = 7.1 Hz, 2 H), 3.45 (s, 2 H), 2.18 (q, *J* = 7.2 Hz, 2 H), 1.45~1.15 (m, 11 H), 0.87 (t, *J* = 6.7 Hz, 3 H).

MS *m/z* (%): 279 [6, $M^+ + 1(^{81}Br)$], 277 [6, $M^+ + 1(^{79}Br)$], 109 (100).

HRMS: calcd for $C_{12}H_{21}{}^{81}BrO_2$: 278.0705. Found: 278.0707, HRMS: calcd for $C_{12}H_{21}{}^{79}BrO_2$: 276.0725 Found: 276.0725.

Ethyl 3-Chloro-3-decenoate (4bc)[24]
Starting from **3b** (105 mg, 0.54 mmol) and $LiCl{\cdot}H_2O$ (97 mg, 1.61 mmol) in CF_3CO_2H (0.50 mL), the mixture was heated under reflux for 2.1 h to afford 104 mg (84%) of **4bc** (*E*/*Z* = 3:1).

(*E*)-**4bc**: liquid.

IR (neat): ν = 2930, 1744, 1368, 1177, 1034 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.80 (t, *J* = 7.3 Hz, 1 H), 4.17 (q, *J* = 7.1 Hz, 2 H), 3.35 (s, 2 H), 2.04 (q, *J* = 7.3Hz, 2 H), 1.45~1.15 (m, 11 H), 0.86 (t, *J* = 6.9 Hz, 3 H), the following data are discernible for the *Z*-isomer: δ = 5.62 (t, *J* = 7.1 Hz, 1 H), 4.17 (q, *J* = 7.1 Hz, 2 H), 3.29 (s, 2 H), 2.18 (q, *J* = 7.1 Hz, 2 H), 1.45~1.15 (m, 11 H), 0.86 (t, *J* = 6.9 Hz, 3 H).

MS *m/z* (%): 235 [55, $M^+(^{37}Cl)$], 233 [19, $M^+(^{35}Cl)$], 197 (100).

2-Methyl-3-iodo-3-butenoate (4ca)[11]
Starting from **3c** (91 mg, 0.72 mmol) and NaI (325 mg, 2.17 mmol) in CF_3CO_2H (1 mL)–HOAc (1 mL), the reaction was carried out at 80 °C for 4 h to afford 173 mg (95%) of **4ca**.

4ca: liquid.

IR (neat): ν = 3400, 2950, 1740, 1615, 1180, 1020 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.22 (d, *J* = 1.5 Hz, 1 H), 5.86 (d, *J* = 1.5 Hz, 1 H), 4.22~4.09 (m, 2 H), 3.27 (q, *J* = 6.7 Hz, 1 H), 1.34~1.21 (m, 6 H).

MS *m/z* (%): 254 (88, M^+), 127 (100).

3-Bromo-2-methyl-3-butenoate (4cb)[11]
Starting from **3c** (108 mg, 0.86 mmol) and $LiBr{\cdot}H_2O$ (180 mg, 1.72 mmol) in CF_3CO_2H (0.5 mL), the mixture was heated under reflux for 40 min to afford 98 mg (8%) of **4cb**.

4cb: liquid.

IR (neat): ν = 3400, 2970, 1740, 1625, 1190, 1020 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.76 (d, *J* = 2.0 Hz, 1 H), 5.57 (d, *J* = 2.0 Hz, 1 H), 4.26~4.12 (m, 2 H), 3.46 (q, *J* = 7.0 Hz, 1 H), 1.36 (d, *J* = 7.0 Hz, 3 H), 1.27 (t, *J* = 7.2Hz, 3 H).

MS *m/z* (%): 207 [4, $M^+(^{81}Br) - 1$], 205 [4, $M^+(^{79}Br) - 1$], 54 (100).

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Ethyl 2-(*n*-Propyl)-3-iodo-3-butenoate (4da)
Starting from **3d** (81 mg, 0.52 mmol) and NaI (157 mg, 1.04 mmol) in CF_3CO_2H (1 mL), the mixture was heated under reflux for 20 min to afford 144 mg (98%) of **4da**.

4da: liquid.

IR (neat): ν = 3410, 2980, 1740, 1620, 1040 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.26 (d, *J* = 1.6 Hz, 1 H), 5.92 (d, *J* = 1.6 Hz, 1 H), 4.30~4.17 (m, 2 H), 3.06 (t, *J* = 7.3 Hz, 1 H), 1.88~1.62 (m, 2 H), 1.44 ~ 1.23 (m, 5 H), 0.92 (t, *J* = 7.1 Hz, 3 H).

MS *m/z* (%): 283 [54, M^+ + 1], 81 (100).

HRMS: calcd for $C_9H_{15}IO_2$: 282.0116. Found: 282.0120.

Ethyl 2-(*n*-Propyl)-3-bromo-3-butenoate (4db)
Starting from **3d** (74 mg, 0.48 mmol) and $LiBr·H_2O$ (100 mg, 0.96 mmol) in CF_3CO_2H (1 mL), the mixture was heated under reflux for 50 min to afford 86 mg (76%) of **4db**.

4db: liquid.

IR (neat): ν = 3420, 2980, 1740, 1630, 1185, 1040 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.77 (d, *J* = 1.8 Hz, 1 H), 5.60 (d, *J* = 1.8 Hz, 1 H), 4.22~4.12 (m, 2 H), 3.30 (t, *J* = 7.5 Hz, 1 H), 1.84~1.74 (m, 2 H), 1.34~1.24 (m, 5 H), 0.90 (t, *J* = 7.5 Hz, 3 H).

MS *m/z* (%): 236 [1, $M^+(^{81}Br)$], 234 [1, $M^+(^{79}Br)$], 81 (100).

HRMS: calcd for $C_9H_{15}{}^{81}BrO_2$: 236.0235. Found: 236.0226, HRMS: calcd for $C_9H_{15}{}^{79}BrO_2$: 234.0256. Found: 234.0277.

Ethyl 2-(*n*-Propyl)-3-chloro-3-butenoate (4dc)
Starting from **3d** (133 mg, 0.86 mmol) and $LiCl·H_2O$ (156 mg, 2.6 mmol) in CF_3CO_2H (1 mL), the mixture was heated under reflux for 10 h to afford 98 mg (63%) of **4dc**.

4dc: liquid.

IR (neat): ν = 3420, 2980, 1740, 1630, 1190, 1040 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.35 (d, *J* = 1.9 Hz,1 H), 5.33 (d, *J* = 1.9 Hz, 1 H), 4.24~4.13 (m, 2 H), 3.29 (t, *J* = 7.5 Hz, 1 H), 1.89~1.73 (m, 2 H), 1.40~1.22 (m, 5 H), 0.93 (t, *J* = 7.5 Hz, 3 H).

MS *m/z* (%): 192 [2, $M^+(^{37}Cl)$], 190 [5, $M^+(^{35}Cl)$], 81 (100).

HRMS: calcd for $C_9H_{15}{}^{35}ClO_2$: 190.0761. Found: 190.0730.

3-Iodo-2-methyl-3-decenoate (4ea)
Starting from **3e** (91 mg, 0.43 mmol) and NaI (91 mg, 0.61 mmol) in CF_3CO_2H (0.5 mL) at r.t., 114 mg (78%) of **4ea** (*E*/*Z* = 3:2) were afforded after 1.1 h.

(*E*)-**4ea**: liquid.

IR: ν = 2900, 1735, 1370, 1180 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.32 (t, *J* = 7.6 Hz, 1 H), 4.25~4.08 (m, 2 H), 3.47 (q, *J* = 6.8 Hz, 1 H), 2.25~2.08 (m, 2 H), 1.50~1.10 (m, 13 H), 0.87 (t, *J* = 6.8 Hz, 3 H), the following data are discernible for *Z*-isomer: δ = 5.69 (t, *J* = 6.9 Hz, 1 H), 4.25~4.08 (m, 2 H), 3.34 (q, *J* = 7.0Hz, 1 H), 2.25~2.08 (m, 2 H), 1.50~1.10 (m, 13 H), 0.87 (t, *J* = 6.8 Hz, 3 H).

MS *m/z* (%): 339 (3, M^+ +1), 81 (100).

HRMS: calcd for $C_{13}H_{23}IO_2$: 338.0742. Found: 338.0751.

Ethyl 2-Methyl-3-bromo-3-decenoate (4eb)
Starting from **3e** (97 mg, 0.65 mmol) and $LiBr·H_2O$ (68 mg, 0.97 mmol) in CF_3CO_2H (0.5 mL) at r.t., 120 mg (89%) of **4eb** (*E*/*Z* = 3:2) were afforded after 3 h.

(*E*)-**4eb**: liquid.

IR (neat): ν = 2929, 2857, 1741, 1194, 1096 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.98 (t, *J* = 7.7 Hz, 1 H), 4.25~4.08 (m, 2 H), 3.76 (q, *J* = 6.9 Hz, 1 H), 2.24~2.04 (m, 2 H), 1.52~1.14 (m, 13 H), 0.88 (t, *J* = 6.6 Hz, 3 H), the following data are discernible for *Z*-isomer: δ = 5.83 (t, *J* = 6.8 Hz, 1 H), 4.25~4.08 (m, 2 H), 3.44(q, *J* = 7.0 Hz, 1 H), 2.24~2.04 (m, 2 H), 1.52~1.14 (m, 13 H), 0.88 (t, *J* = 6.6 Hz, 3 H).

MS *m/z* (%): 291 [6, M^+ (^{79}Br)], 293 [6, M^+ (^{81}Br)], 41 (100).

HRMS: calcd for $C_{13}H_{23}{}^{81}BrO_2$: 293.0940. Found: 293.0955, HRMS: calcd for $C_{13}H_{23}{}^{79}BrO_2$: 291.0960. Found: 291.0952.

Ethyl 2-Methyl-3-chloro-3-decenoate (4ec) and 3-Methyl-5-hexyl-2(5*H*)-furanone (6e)[25]
Starting from **3e** (117 mg, 0.56 mmol) and $LiCl·H_2O$ (168 mg, 2.78 mmol) in CF_3CO_2H (0.5 mL), the reaction was carried out at 80 °C for 17 h to afford 87 mg (63%) of **4ec** (*E*/*Z* = 3:2) and 19.1 mg (19%) of **6e**.

(*E*)-**4ec**: liquid.

IR (neat): ν = 2900, 1740, 1370, 1180 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.71 (t, *J* = 7.8 Hz, 1 H), 4.20~4.05 (m, 2 H), 3.76 (q, *J* = 6.9 Hz, 1 H), 2.15~2.05 (m, 2 H), 1.60~1.15 (m, 13 H), 0.87 (t, *J* = 6.0 Hz, 3 H), the following data are discernible for the *Z*-isomer: δ = 5.63 (t, *J* = 7.0 Hz, 1 H), 4.20~4.05 (m, 2 H), 3.38 (q, *J* = 7.1Hz, 1 H), 2.15~2.05 (m, 2 H), 1.60~1.15 (m, 13 H), 0.87 (t, *J* = 6.0 Hz, 3 H).

MS *m/z* (%): 247 [15, M^+ (^{35}Cl)], 249 [5, M^+ (^{37}Cl)], 67 (100).

HRMS: calcd for $C_{13}H_{23}{}^{37}ClO_2$: 248.1357. Found: 248.1365, HRMS: calcd for $C_{13}H_{23}{}^{35}ClO_2$: 246.1387. Found: 246.1385.

6e[25]: solid.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.00 (br s, 1 H), 4.91~4.81 (m, 1 H), 1.90 (s, 3 H), 1.80~1.10 (m, 10 H), 0.86 (t, *J* = 6.5 Hz, 3 H).

MS *m/z* (%): 183 (M^+ + 1), 41 (100).

Ethyl 2-Methyl-3-iodo-3-pentenoate (4fa)
Starting from **3f** (70 mg, 0.50 mmol) and NaI (112 mg, 0.75 mmol) in CF_3CO_2H (0.5 mL) at r.t., 104 mg (78%) of **4fa** (*E*/*Z* = 1:1) were afforded after 50 min.

(*E*)-**4fa**: liquid.

IR (neat): ν = 2960, 1740, 1370, 1190 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.38 (q, *J* = 7.0 Hz, 1 H), 4.20~4.05 (m, 2 H), 3.49 (q, *J* = 6.5 Hz, 1 H), 1.71 (d, *J* = 7.0 Hz, 3 H), 1.35~1.15 (m, 6 H), the following data are discernible for the *Z*-isomer: δ = 5.79 (q, *J* = 6.5 Hz, 1 H), 4.20~4.05 (m, 2 H), 3.38 (q, *J* = 6.9 Hz, 1 H), 1.76 (d, *J* = 6.5 Hz, 3 H), 1.35~1.15 (m, 6 H).

MS *m/z* (%): 269 (38, M^+ + 1), 268 (12, M^+), 67 (100).

HRMS: calcd for $C_8H_{13}IO_2$: 267.9959. Found: 267.9953.

Ethyl 2-Methyl-3-bromo-3-pentenoate (4fb)
Starting from **3f** (80 mg, 0.57 mmol) and $LiBr·H_2O$ (90 mg, 0.86 mmol) in CF_3CO_2H (0.5 mL) at r.t., 94 mg (74%) of **4fb** (*E*/*Z* = 1.2:1) were afforded after 2.5 h.

(*E*)-**4fb**: liquid.

IR (neat): ν = 2985, 1739, 1195 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.04 (q, *J* = 7.1 Hz, 1 H), 4.25~4.10 (m, 2 H), 3.77 (q, *J* = 6.9 Hz, 1 H), 1.71 (d, *J* = 7.0 Hz, 3 H), 1.35~1.15 (m, 6 H), the following data are discernible for the *Z*-isomer: δ = 5.92 (q, *J* = 6.4 Hz, 1 H), 4.25~4.15 (m, 2 H), 3.46 (q, *J* = 6.9 Hz, 1 H), 1.76 (d, *J* = 6.4 Hz, 3 H), 1.35~1.15 (m, 6 H).

MS *m/z* (%): 223 [22, $M^+(^{81}Br)$], 221 [22, $M^+(^{79}Br)$], 67 (100).

HRMS: calcd for $C_8H_{13}{}^{81}BrO_2$: 222.0079. Found: 222.0074, HRMS: calcd for $C_8H_{13}{}^{79}BrO_2$: 220.0099. Found: 220.0086.

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Ethyl 2-Methyl-3-chloro-3-pentenoate (4fc)
Starting from **3f** (100 mg, 0.71 mmol) and LiCl·H_2O (186 mg, 2.86 mmol) in CF_3CO_2H (0.5 mL), the mixture was heated under reflux for 40 min to afford 69 mg (5%) of **4fc** (*E*/*Z* = 1.5:1).

(*E*)-**4fc**: liquid.

IR (neat): ν = 2980, 1740, 1450 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.77 (q, *J* = 7.0 Hz, 1 H), 4.21~4.07 (m, 2 H), 3.76 (q, *J* = 6.9 Hz, 1 H), 1.71 (d, *J* = 7.2 Hz, 3 H), 1.38~1.18 (m, 6 H), the following data are discernible for the *Z*-isomer: δ = 5.70 (q, *J* = 7.0 Hz, 1 H), 4.21~4.07 (m, 2 H), 3.38 (q, *J* = 7.0 Hz, 1 H), 1.75 (d, *J* = 6.6 Hz, 3 H), 1.38~1.18 (m, 6 H).

MS *m/z* (%): 178 [4.8, $M^+(^{37}Cl)$], 176 [13.5, $M^+(^{35}Cl)$], 67 (100).

HRMS: calcd for $C_8H_{13}{}^{37}ClO_2$: 178.0575. Found: 178.0853, HRMS: calcd for $C_8H_{13}{}^{35}ClO_2$: 176.0604. Found: 176.0597.

Ethyl 4-Methyl-3-iodo-3-pentenoate (4ga) and 5,5-Dimethyl-2-(5*H*)-furanone (6g)[26]
Starting from **3g** (65 mg, 0.46 mmol) and NaI (139 mg, 0.93 mmol) in HOAc (0.5 mL), the reaction was carried out at 80 °C for 4 h to afford 28.6 mg (23%) of **4ga** and 7.7 mg (15%) of **6g**.

4ga: liquid.

IR: ν = 2979, 1732, 1328, 1180 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 4.18 (q, *J* = 7.1 Hz, 2 H), 3.73 (s, 2 H), 2.00 (s, 3H), 1.87 (s, 3 H), 1.28 (t, *J* = 7.1 Hz, 3 H).

MS *m/z* (%): 269 (37, $M^+ + 1$), 268 (50, M^+), 67 (100).

HRMS: calcd for $C_8H_{13}IO_2$: 267.9959. Found: 267.9953.

6g: solid.

IR: ν = 2983, 1755, 1132 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.46 (d, *J* = 5.5Hz, 1 H), 6.04 (d, *J* = 5.5 Hz, 1 H), 1.54 (s, 6 H).

MS *m/z* (%): 112 (10, M^+), 97(100).

Ethyl 4-Methyl-3-bromo-3-pentenoate (4gb) and 5,5-Dimethyl-2-(5*H*)-furanone (6g)[26]
Starting from **3g** (101 mg, 0.72 mmol) and LiBr·H_2O (152 mg, 1.44 mmol) in HOAc (0.5 mL), the reaction was carried out at 80 °C for 15 h to afford 36.6 mg (23%) of **4ga** and 33.7 mg (42%) of **6g**.

4gb: liquid.

IR (neat): ν = 2920, 1742, 1178 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 4.17 (q, *J* = 7.1 Hz, 2 H), 3.56 (s, 2 H), 1.91 (s, 3 H), 1.79 (s, 3 H), 1.26 (t, *J* = 7.1 Hz, 3 H).

MS *m/z* (%): 223 [33, $M^+ + 1(^{81}Br)$], 221 [34.5, $M^+ + 1(^{79}Br)$], 67 (100).

HRMS: calcd for $C_8H_{13}{}^{81}BrO_2$: 222.0079. Found: 222.0078, HRMS: calcd for $C_8H_{13}{}^{79}BrO_2$: 220.0099. Found: 220.0100.

3,5,5-Trimethyl-2-(5*H*)-furanone (6h)[27]
Starting from **3h** (63 mg, 0.41 mmol) and LiBr·H_2O (86 mg, 0.82 mmol) in HOAc (0.25 mL)–CF_3CO_2H (0.25 mL), the reaction was carried out at 80 °C for 10 min to afford 35 mg (67%) of **6h**.

6h: solid.

IR (KBr): ν = 2983, 1745, 1300, 1082 cm^{-1}.

^1H NMR (300MHz, $CDCl_3$): δ = 6.96 (q, *J* = 1.3 Hz, 1 H), 1.85 (d, *J* = 1.3 Hz, 3 H), 1.41 (s, 6 H).

MS *m/z* (%): 126 (7, M^+), 43 (100).

Pd(0)-catalyzed Coupling Reactions of Ethyl 3-iodo-3-alkenoate (4ca)

Ethyl 2-Methyl-3-methylene-2-heptanoate (7a)
To a solution of dried $ZnBr_2$ (510 mg, 2.27 mmol) in THF (3 mL) was added *n*-BuLi (2 M, 1 mL) at –78 °C. After stirring at this temperature for 20 min, the solution was warmed up to r.t. and stirred for additional 30 min. Then the *n*-BuZnBr solution was added at r.t. to a solution of ethyl 2-methyl-3-iodo-3-butenoate (**4ca**) (111 mg, 0.44 mmol) and $Pd(PPh_3)_4$ (25 mg, 0.022 mmol) in THF (1 mL) under Ar. After 3 h, the solvent was removed by evaporation. Chromatography on silica gel (eluent: petroleum ether–ethyl acetate, 50:1) afforded 55.1 mg (90%) of **7a** as a liquid.

7a: liquid

IR (neat): ν = 1735, 1640 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 4.90 (s, 1 H), 4.86 (s, 1 H), 4.12 (q, *J* = 7.1 Hz, 2 H), 3.12 (q, *J* = 7.1 Hz, 1 H), 2.04 (q, *J* = 4.3 Hz, 2 H), 1.42~1.20 (m, 10 H), 0.89 (t, *J* = 7.1 Hz, 3 H).

MS *m/z* (%): 184 (1, M^+), 69 (100).

HRMS: calcd for $C_{11}H_{20}O_2$: 184.1463. Found: 184.1468.

Ethyl 2-Methyl-3-phenyl-3-butenoate (7b)[17,18]
To a solution of PhI (125.6 mg, 0.62 mmol) in THF (1 mL) was added *n*-BuLi (2.0 M, 0.31 mL) at –78 °C while stirring. After the addition, the reaction mixture was warmed up to –40 °C, and a solution of dried $ZnBr_2$ (680 mg, 3 mmol) in THF (2 mL) was added. To this solution was added a solution of ethyl 2-methyl-3-iodo-3-butenoate (**4ca**) (113 mg, 0.44 mmol) and $Pd(PPh_3)_4$ (26 mg, 0.02 mmol) in THF (1 mL) under Ar. After 6 h, the solvent was removed by evaporation. Chromatography on silica gel (eluent: petroleum ether–ethyl acetate, 60:1) afforded 88.3 mg (70%) of **7b**.

7b: liquid

IR (neat): ν = 3400, 2960, 1735, 1180, 1020 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.45~7.23 (m, 5 H), 5.39 (s, 1 H), 5.23 (s, 1 H), 4.11 (q, *J* = 7.1 Hz, 2 H), 3.68 (q, *J* = 7.1 Hz, 1 H), 1.39 (d, *J* = 7.1 Hz, 3 H), 1.16 (t, *J* = 7.2 Hz, 3 H).

MS *m/z* (%): 204 (19, M^+), 91 (100).

Ethyl 2-Methyl-3-methylene-5-phenyl-4-pentynoate (7c)
To a Schlenk reaction tube were added ethyl 2-methyl-3-iodo-3-butenoate (**4ca**) (88 mg, 0.35 mmol), MeCN (0.5 mL), $PdCl_2(PPh_3)_2$ (15 mg, 0.019 mmol), CuI (10.9 mg, 0.057 mmol), Et_3N (0.08 mL, 0.52 mmol), phenylacetylene (0.05 mL, 0.46 mmol), and MeCN (0.5 mL). This reaction mixture was stirred at r.t. for 6.5 h, then the solvent was removed by evaporation. Chromatography on silica gel (eluent: petroleum ether–ethyl acetate, 50:1) afforded 67 mg (85%) of **7c**.

7c: liquid

IR (neat): ν = 2359, 1735, 1600 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.44~7.41 (m, 2 H), 7.31~7.28 (m, 3 H), 5.57 (s, 1 H), 5.46 (s, 1 H), 4.22~4.12 (m, 2 H), 3.33 (q, *J* = 7.0 Hz, 1 H), 1.42 (d, *J* = 7.0 Hz, 3 H), 1.27 (t, *J* = 7.0 Hz, 3 H).

MS *m/z* (%): 228 (57, M^+), 156 (100).

HRMS: calcd for $C_{15}H_{16}O_2$: 228.1150. Found: 228.1145.

Ethyl 3-Phenyl-3-pentenoate (7d)
To a solution of PhI (153 mg, 0.75 mmol) in THF (1 mL) was added *n*-BuLi (2.0 M, 0.38 mL) at –78 °C while stirring. After the addition, the reaction mixture was warmed up to –40 °C, and a solution of dried $ZnBr_2$ (680 mg, 3 mmol) in THF (2 mL) was added. To this solution was added a solution of ethyl 3-iodo-3-pentenoate **(4aa)** (*E*/*Z* = 2.5:1) (127 mg, 0.5 mmol) and $Pd(PPh_3)_4$ (28 mg, 0.038

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mmol) in THF (1 mL) under Ar. After 8 h, chromatography on silica gel (eluent: petroleum ether–ethyl acetate, 60:1) afforded 77 mg (76%) of (*E*)-**7d** (*E*/*Z* = 2.5:1) as a liquid.

IR (neat): ν = 2985, 1738, 1155 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.37~7.18 (m, 5 H), 6.03 (q, *J* = 6.8 Hz, 1 H), 4.00~4.30 (m, 2 H), 3.30 (s, 2 H), 1.83 (q, *J* = 6.8 Hz, 3 H), 1.40~1.10 (m, 3 H), the following data were discernible for *Z*-isomer: d = 7.37~7.18 (m, 5 H), 5.71 (q, *J* = 6.7 Hz, 1 H), 4.00~4.30 (m, 2 H), 3.33 (s, 2 H), 1.62 (q, *J* = 6.7 Hz, 3 H), 1.40~1.10 (m, 3 H).

MS *m/z* (%): 204 (100, M^+).

HRMS: calcd for $C_{13}H_{16}O_2$: 204.1150. Found: 204.1193.

Ethyl 3-Ethylidene-5-phenyl-4-pentynoate (7e)

To a Schlenk reaction tube were added ethyl 3-iodo-3-pentenoate **(4aa)** (*E*/*Z* = 2.5:1) (256 mg, 1.01 mmol), MeCN (0.5 mL), $PdCl_2(PPh_3)_2$ (41 mg, 0.05 mmol), CuI (19 mg, 0.1 mmol), Et_3N (0.2 mL, 1.5 mmol), phenylacetylene (0.2 mL, 1.8 mmol), and MeCN (0.5 mL) under Ar. This reaction mixture was stirred at r.t. for 6 h, then the solvent was removed by evaporation. Chromatography on silica gel (eluent: petroleum ether–ethyl acetate, 50:1) afforded 188 mg (82%) of (*E*)-**7e** (*E*/*Z* = 2.5:1).

(*E*)-7e: liquid

IR (neat): ν = 2985, 1740, 1259, 1181, 1034 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.44~7.38 (m, 2 H), 7.28~7.24 (m, 1 H), 6.22 (q, *J* = 7.1 Hz, 1 H), 4.35~4.10 (m, 2 H), 3.26 (s, 2 H), 1.78 (d, *J* = 7.1 Hz, 3 H), 1.26 (t, *J* = 7.1 Hz, 3 H), the following data were discernible for *Z*-isomer: δ = 7.44~7.38 (m, 2 H), 7.28~7.24 (m, 1 H), 5.93 (q, *J* = 7.0 Hz, 1 H), 4.35~4.10 (m, 2 H), 3.19 (s, 2 H), 1.88 (d, *J* = 7.0 Hz, 3 H), 1.26 (t, *J* = 7.1 Hz, 3 H).

MS *m/z* (%): 228 (100, M^+).

HRMS: calcd for $C_{15}H_{16}O_2$: 228.1150. Found: 228.1133.

Hydrohalogenation Reaction of 2,3-Dienamides

3-Iodo-4-methyl-3-pentenamide (9aa)

Starting from **8a** (50 mg, 0.45 mmol) and NaI (101 mg, 0.67 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 4 h to afford 94 mg (87%) of **9aa**.

White solid; mp 153–154 °C ($CHCl_3$).

IR (KBr): ν = 3353, 3180, 1666, 1628 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.76 (br s, 2 H), 3.66 (s, 2 H), 2.00 (s, 3 H), 1.88 (s, 3 H).

MS *m/z* (%): 239 (10.95, M^+), 112 (100).

Anal. Calcd for $C_6H_{10}INO$: C, 30.15; H, 4.22; N, 5.86. Found: C, 29.90; H, 4.17; N, 5.77.

3-Bromo-4-methyl-3-pentenamide (9ab)

Starting from **8a** (30 mg, 0.27 mmol) and $LiBr·H_2O$ (43 mg, 0.41 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 4 h to afford 50 mg (96%) of **9ab**.

White solid; mp 145–146 °C ($CHCl_3$).

IR (KBr): ν = 3354, 3180, 1666, 1628 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.78 (br s, 1 H), 5.74 (br s, 1 H), 3.53 (s, 2 H), 1.95 (s, 3 H), 1.84 (s, 3 H).

MS *m/z* (%): 193 [16.69, $M^+(^{81}Br)$] 191 [12.88, $M^+(^{79}Br)$], 112 (100).

Anal. Calcd for $C_6H_{10}BrNO$: C, 37.52; H, 5.25; N, 7.29. Found: C, 37.21; H, 5.19; N, 7.09.

3-Chloro-4-methyl-3-pentenamide (9ac)

Starting from **8a** (30 mg, 0.27 mmol) and $LiCl·H_2O$ (25 mg, 0.41 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 7 h to afford 30 mg (75%) of **9ac**.

White solid; mp 130 °C ($CHCl_3$, sublimation).

IR (KBr): ν = 3354, 3180, 1666, 1630 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.78 (br s, 2 H), 3.37 (s, 2 H), 1.91 (s, 3 H), 1.83 (s, 3 H).

MS *m/z* (%): 149 [18.72, $M^+(^{37}Cl)$], 147 [48.07, $M^+(^{35}Cl)$], 67 (100).

Anal. Calcd for $C_6H_{10}ClNO$: C, 48.82; H, 6.83; N, 9.49. Found: C, 48.77; H, 6.70; N, 9.48.

***N*-*t*-Butyl-3-iodo-3-octenamide (9ba)**

Starting from **8b** (27 mg, 0.14 mmol) and NaI (35 mg, 0.23 mmol) in HOAc (1 mL) at r.t., 16 mg (36 %) of (*E*)-**9ba** and 28 mg (62%) of (*Z*)-**9ba** were afforded after 4 h.

(*E*)-**9ba**

White solid; mp 53–54 °C (*n*-hexane).

IR (KBr): ν = 3305, 1647, 1548 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.41 (t, *J* = 7.59 Hz, 1 H), 5.54 (br s, 1 H), 3.40 (s, 2 H), 2.07 (q, *J* = 7.22 Hz, 2 H), 1.43–1.20 (m, 13 H), 0.88 (t, *J* = 7.03 Hz, 3 H).

MS *m/z* (%): 324 (62,97, M^+ + 1), 57 (100).

HRMS: calcd for $C_{12}H_{22}INO$: 323.0745. Found: 323.0746.

(*Z*)-**9ba**

White solid; mp 47.5–49 °C (*n*-hexane).

IR (KBr): ν = 3294, 1645, 1550 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.67 (t, *J* = 6.48 Hz, 1 H), 5.46 (br s, 1 H), 3.41 (s, 2 H), 2.15 (q, *J* = 6.48 Hz, 2 H), 1.45–1.24 (m, 13 H), 0.90 (t, *J* = 6.77 Hz, 3 H).

MS *m/z* (%): 324 (36,27, M^+ + 1), 57 (100).

HRMS: calcd for $C_{12}H_{22}INO$: 323.0745. Found: 323.0739.

Starting from **8b** (32 mg, 0.16 mmol) and NaI (36 mg, 0.24 mmol) in CF_3CO_2H (1 mL) at r.t., 34 mg (64%) of **(***E***)-9ba** and 18 mg (34%) of (*Z*)-**9ba** were afforded after 0.5 h.

***N*-*t*-Butyl-3-bromo-3-octenamide (9bb)**

Starting from **8b** (56 mg, 0.29 mmol) and $LiBr·H_2O$ (45 mg, 0.43 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 22 h to afford 27 mg (34%) of **(***E***)-9bb** and 44 mg (56%) of (*Z*)-**9bb**.

(*E*)-**9bb**

White solid; mp 49–50 °C (*n*-hexane).

IR (KBr): ν = 3290, 1646, 1556 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.10 (t, *J* = 7.74 Hz, 1 H), 5.58 (br s, 1 H), 3.34 (s, 2 H), 2.07 (q, *J* = 7.10 Hz, 2 H), 1.40–1.30 (m, 13 H), 0.90 (t, *J* = 7.04 Hz, 3 H).

MS *m/z* (%): 278 [5.01, M^+ + 1(^{81}Br)], 276 [4.93, M^+ + 1(^{79}Br)], 57 (100).

HRMS: calcd for $C_{12}H_{22}{}^{81}BrNO$: 277.0865. Found: 277.0869.

(*Z*)-**9bb**

White solid; mp 53–54 °C (*n*-hexane).

IR (KBr): ν = 3294, 1647, 1551 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.85 (t, *J* = 6.78 Hz, 1 H), 5.55 (br s, 1 H), 3.28 (s, 2 H), 2.20 (q, *J* = 6.70 Hz, 2 H), 1.43–1.34 (m, 13 H), 0.91 (t, *J* = 7.04 Hz, 3 H).

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MS *m/z* (%): 278 (11.06, $M^+ + 1(^{81}Br)$], 276 [11.07, $M^+ + 1(^{79}Br)$], 57 (100).

HRMS: calcd for $C_{12}H_{22}{}^{81}BrO$: 277.0865. Found: 277.0864.

Starting from **8b** (37 mg, 0.19 mmol) and $LiBr{\cdot}H_2O$ (30 mg, 0.29 mmol) in CF_3CO_2H (1 mL) at 70 °C, 25 mg (48%) of **(***E***)-9bb** and 11 mg (21%) of **(***Z***)-9bb** were afforded after 4 h.

***N*-*t*-Butyl-3-chloro-3-octenamide (9bc)**
Starting from **8b** (66 mg, 0.34 mmol) and $LiCl{\cdot}H_2O$ (31 mg, 0.51 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 22 h to afford 44 mg (56%) of **(***E***)-9bc** and 18 mg (23%) of (*Z*)-**9bc**.

(*E*)-**9bc**

White solid; mp 48–49 °C (*n*-hexane).

IR (KBr): ν = 3289, 1646, 1558 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.84 (t, *J* = 7.75 Hz, 1 H), 5.60 (br s, 1 H), 3.21 (s, 2 H), 2.07 (q, *J* = 7.22 Hz, 2 H), 1.38–1.31 (m, 13 H), 0.89 (t, *J* = 7.05 Hz, 3 H).

MS *m/z* (%): 234 [0.44, $M^+ + 1(^{37}Cl)$], 232 [1.33, $M^+ + 1(^{35}Cl)$], 57 (100).

HRMS: calcd for $C_{12}H_{22}{}^{35}ClNO$: 231.1390. Found: 231.1391.

(*Z*)-**9bc**

White solid; mp 52–54 °C (*n*-hexane).

IR (KBr): ν = 3293, 1647, 1555 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.67 (t, *J* = 7.03 Hz, 1 H), 5.55 (br s, 1 H), 3.14 (s, 2 H), 2.22 (q, *J* = 7.08 Hz, 2 H), 1.42–1.33 (m, 13 H), 0.91 (t, *J* = 7.02 Hz, 3 H).

MS *m/z* (%): 234 [0.66, $M^+ + 1(^{37}Cl)$], 232 [2.03, $M^+ + 1(^{35}Cl)$], 57 (100).

HRMS: calcd for $C_{12}H_{22}{}^{35}ClNO$: 231.1390. Found: 231.1412.

Starting from **8b** (66 mg, 0.34 mmol) and $LiCl{\cdot}H_2O$ (31 mg, 0.51 mmol) in CF_3CO_2H (1 mL) at 70 °C, 12 mg (15%) of **(***E***)-9bc** and 6 mg (8%) of **(***Z***)-9bc** were afforded after 26 h.

***N*-*t*-Butyl-3-iodo-3-butenamide (9ca)**
Starting from **8c** (50 mg, 0.36 mmol) and NaI (81 mg, 0.54 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 18 h to afford 93 mg (97%) of **9ca**.

White solid; mp 89–90 °C (*n*-hexane).

IR (KBR): ν = 3300, 1640, 1550 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.19 (s, 1 H), 5.89 (s**,** 1 H), 5.48 (br s, 1 H), 3.34 (s, 2 H), 1.35 (s, 9 H).

MS *m/z* (%): 268 (29.45, $M^+ + 1$), 57 (100).

Anal. Calcd for $C_8H_{14}INO$: C, 35.97; H, 5.28; N, 5.24. Found: C, 35.84; H, 5.23; N, 5.20.

***N*-*t*-Butyl-3-bromo-3-butenamide (9cb)**
Starting from **8c** (80 mg, 0.58 mmol) and $LiBr{\cdot}H_2O$ (91 mg, 0.87 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 23 h to afford 118 mg (93%) of **9cb**.

White solid; mp 83.5–84.5 °C (*n*-hexane).

IR (KBr): ν = 3300, 1640, 1545 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.78 (s, 1 H), 5.62 (s, 1 H), 5.56 (br s, 1 H), 3.30 (s, 2 H), 1.36 (s, 9 H).

MS *m/z* (%): 222 [1.37, $M^+ + 1(^{81}Br)$], 220 [1.45, $M^+ + 1(^{79}Br)$], 57 (100).

Anal. Calcd for $C_8H_{14}BrNO$: C, 43.65; H, 6.41; N, 6.36. Found: C, 43.66; H, 6.49; N, 6.10.

***N*-*t*-Butyl-3-chloro-3-butenamide (9cc)**
Starting from **8c** (112 mg, 0.81 mmol) and $LiCl{\cdot}H_2O$ (74 mg, 1.2 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 23 h to afford 128 mg (91%) of **9cc**.

White solid; mp 83–84 °C (*n*-hexane).

IR (KBr): ν = 3300, 1640, 1545 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.60 (br s, 1 H), 5.35 (s, 1 H), 5.34 (s, 1 H), 3.16 (s, 2 H), 1.34 (s, 9 H).

MS *m/z* (%): 178 [37.64, $M^+ + 1(^{37}Cl)$], 176 [100, $M^+ + 1(^{35}Cl)$].

Anal. Calcd for $C_8H_{14}ClNO$: C, 54.70; H, 8.03; N, 7.97. Found: C, 54.87; H, 8.16; N, 7.81.

***N*-Benzyl-3-iodo-2-methyl-3-butenamide (9da)**
Starting from **8d** (95 mg, 0.51 mmol) and NaI (114 mg, 0.76 mmol) in CF_3CO_2H (1 mL) at r.t., 155 mg (97%) of **9da** were afforded after 1 h.

White solid; mp 83–84 °C (*n*-hexane).

IR (film): ν = 3280, 1640, 1535 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.32–7.15 (m, 5 H), 6.30 (s,1 H), 5.98 (br s, 1 H), 5.93 (s, 1 H), 4.53–4.40 (m, 2 H), 3.15 (q, *J* = 6.89 Hz, 1 H), 1.30 (d, *J* = 7.09 Hz, 3 H).

MS *m/z* (%): 316 (7.82, $M^+ + 1$), 91 (100).

Anal. Calcd for $C_{12}H_{14}INO$: C, 45.73; H, 4.48; N, 4.44. found: C, 45.76; H, 4.41; N, 4.38.

Starting from **8d** (31 mg, 0.17 mmol) and aq HI (45 %, 0.5 mL) at r.t., 40 mg (77%) of **9da** were afforded after 11 h.

***N*-Benzyl-3-bromo-2-methyl-3-butenamide (9db)**
Starting from **8d** (100 mg, 0.53 mmol) and $LiBr{\cdot}H_2O$ (84 mg, 0.8 mmol) in CF_3CO_2H (1 mL) at r.t., 142 mg (91%) of **9db** were afforded after 24 h.

White solid; mp 71–72 °C (*n*-hexane).

IR (KBr): ν = 3280, 1640, 1545 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.36–7.25 (m, 5 H), 6.01 (br s, 1 H), 5.84 (s, 1 H), 5.63 (s, 1 H), 4.53–4.40 (m, 2 H), 3.37 (q, *J* = 6.92 Hz, 1 H), 1.41 (d, *J* = 6.92 Hz, 3 H).

MS *m/z* (%): 270 [18.21, $M^+ + 1(^{81}Br)$], 268 [18.32, $M^+ + 1(^{79}Br)$], 91 (100).

Anal. Calcd for $C_{12}H_{14}BrNO$: C, 53.75; H,5.26; N, 5.22. Found: C, 53.86; H, 5.23; N, 5.13.

Starting from **8d** (31 mg, 0.17 mmol) and aq HBr (40 %, 0.5 mL) at r.t., 38 mg (86%) of **9db** were afforded after 11 h.

***N*-Benzyl-3-chloro-2-methyl-3-butenamide (9dc)**
Starting from **8d** (100 mg, 0.53 mmol) and $LiCl{\cdot}H_2O$ (49 mg, 0.81 mmol) in CF_3CO_2H (1 mL), the reaction was carried out at 40 °C for 70 h to afford 104 mg (87%) of **9dc**.

White solid; mp 62.5–63.5 °C (*n*-hexane).

IR (KBr): ν = 3280, 1640, 1550 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.37–7.27 (m, 5 H), 6.05 (br s, 1 H), 5.40 (s, 1 H), 5.38 (s, 1 H), 4.55–4.41 (m, 2 H), 3.34 (q, *J* = 6.98 Hz, 1 H), 1.44 (d, *J* = 7.02 Hz, 3 H).

MS *m/z* (%): 224 [0,72, $M^+ + 1(^{35}Cl)$], 226 [0.23, $M^+ + 1(^{37}Cl)$], 91 (100).

Anal. Calcd for $C_{12}H_{14}ClNO$: C, 64.43; H, 6.31; N, 6.26. Found: C, 64.56; H, 6.41; N, 6.50.

Starting from **8d** (33 mg, 0.18 mmol) and aq HCl (37%, 0.5 mL) at r.t. for 11 h, 37 mg (94%) of **9dc** were afforded.

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Pd(0)-catalyzed Coupling Reaction of *N*-*t*-Butyl-3-iodo-3-butenamide (9ca)

***N*-*t*-Butyl-3-phenyl-3-butenamide (7f)**
To a solution of PhI (219 mg, 1.07 mmol) in THF (1 mL) was added *n*-BuLi (2.0 M, 0.54 mL) at –78 °C while stirring. After the addition, the reaction mixture was warmed up to –40 °C, and a solution of dried $ZnBr_2$ (602 mg, 2.68 mmol) in THF (2 mL) was added. To this solution was added a solution of *N*-*t*-butyl-3-iodo-3-butenamide (**9ca)** (143 mg, 0.54 mmol) and $Pd(PPh_3)_4$ (31 mg, 0.027mmol) in THF (1 mL). After stirring at r.t. for 8 h, the solvent was removed by evaporation. Chromatography on silica gel afforded 101 mg (87%) of **7f**.

Solid; mp 108–110 °C (*n*-hexane).

IR (KBr): ν = 3289, 1641, 1555, 898, 779, 705 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.46–7.32 (m, 5 H), 5.62 (s, 1 H), 5.52 (br s, 1 H), 5.26 (s, 1 H), 3.38 (s, 2 H), 1.22 (s, 9 H).

MS *m/z* (%): 217 (38, M^+), 57 (100).

HRMS: calcd for $C_{14}H_{19}NO$: 217.14667. Found: 217.14800.

***N*-t-Butyl-3-methylene-5-phenyl-4-pentynamide (7g)**
To a Schlenk reaction tube were added under Ar *N*-*t*-butyl-3-iodo-3-butenamide (**9ca)** (94 mg, 0.35 mmol), MeCN (1 mL), $PdCl_2(PPh_3)_2$ (14 mg, 0.018 mmol), CuI (7 mg, 0.035 mmol), Et_3N (53 mg, 0.53 mmol), phenylacetylene (47 mg, 0.46 mmol), and MeCN (1 mL). This reaction mixture was stirred at r.t. for 19 h to afford 70 mg (83%) of **7g**.

Solid; mp 77–78 °C(*n*-hexane).

IR (film): ν = 3302, 1647, 1556, 913, 751, 689 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.42–7.29 (m, 5 H), 5.75 (br s, 1 H), 5.59 (d, *J* = 0.95 Hz, 1 H), 5.49 (d, *J* = 0.95 Hz, 1 H), 3.08 (s, 2 H), 1.32 (s, 9 H).

MS *m/z* (%): 241 (27, M^+), 69 (100).

HRMS: calcd for $C_{16}H_{19}NO$: 241.14667. Found: 241.14614.

Hydrohalogenation Reaction of 2,3-Dienenitriles

3-Bromo-3-butenenitrile (11ab)[28]
Starting from **10a** (131 mg, 2.0 mmol) and $LiBr{\cdot}H_2O$ (231 mg, 2.2 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 21 h to afford 87% (NMR yield) of **11ab** as a liquid.

IR (neat): ν = 2254, 1632 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.97 (t, *J* = 1.30 Hz, 1 H), 5.64 (t, *J* = 1.33 Hz, 1 H), 3.45 (d, *J* = 0.98 Hz, 2 H).

MS *m/z* (%): 147 [68.76, $M^+(^{81}Br)$], 145 [65.97, M^+ (^{79}Br)], 66 (100).

3-Chloro-3-butenenitrile (11ac)[29]
Starting from **10a** (130 mg, 2.0 mmol) and $LiCl{\cdot}H_2O$ (133 mg, 2.2 mmol) in HOAc (1 mL), the reaction was carried out at 70 °C for 24 h to afford 78% (NMR yield) of **11ac** as a liquid.

IR (neat): ν = 2256, 1636 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.63 (t, *J* = 1.20 Hz, 1 H), 5.49 (t, *J* = 1.06 Hz, 1 H), 3.43 (d, *J* = 1.04 Hz, 2 H).

MS *m/z* (%): 103 [19.89, $M^+(^{37}Cl)$], 101 [58.99, $M^+(^{35}Cl)$], 66 (100).

3-Iodo-4-methyl-3-pentenenitrile (11ba)
Starting from **10b** (93 mg, 1.0 mmol) and NaI (225 mg, 1.5 mmol) in CF_3CO_2H (0.5 mL) and HOAc (0.5 mL), the reaction was carried out at 70 °C for 1.5 h to afford 176 mg (80%) of **11ba** as a liquid.

IR (neat): ν = 2250, 1640 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 3.75 (s, 2 H) 1.98 (s, 3 H) 1.91 (s, 3 H).

MS *m/z* (%): 221 (19.94, M^+), 94 (100).

HRMS: calcd for C_6H_8IN: 220.9700. Found: 220.9684.

Starting from **10b** (96 mg, 1.0 mmol) and NaI (165 mg, 1.1 mmol) in HOAc (1.0 mL) at 70 °C, 84 mg (37%) of **11ba** was afforded after 12 h.

3-Bromo-4-methyl-3-pentenenitrile (11bb)
Starting from **10b** (93 mg, 1.0 mmol) and $LiBr{\cdot}H_2O$ (210 mg, 2 mmol) in HOAc (0.5 mL) and CF_3CO_2H (0.5 mL), the reaction was carried out at 70 °C for 7.5 h to afford 135 mg (78%) of **11bb** as a liquid.

IR (neat): ν = 2240,1660 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 3.61 (s, 2 H) 1.92 (s, 3 H) 1.85 (s, 3 H).

MS *m/z* (%): 175 [26.78, $M^+(^{81}Br)$], 173 [24.19, $M^+(^{79}Br)$], 67 (100).

HRMS: calcd for $C_6H_8{}^{79}BrN$: 172.9824. Found: 172.9840, HRMS: calcd for $C_6H_8{}^{81}BrN$: 174.9820. Found: 174.9798.

Starting from **10b** (93 mg, 1.0 mmol) and $LiBr{\cdot}H_2O$ (525 mg, 5 mmol) in HOAc (1.0 mL) at 85 °C, 30 mg (17%) of **11bb** was afforded after 7.5 h.

3-Chloro-4-methyl-3-pentenenitrile (11bc)
Starting from **10b** (93 mg, 1.0 mmol) and $LiCl{\cdot}H_2O$ (121 mg, 2 mmol) in HOAc (0.5 mL) and CF_3CO_2H (0.5 mL), the reaction was carried out at 70 °C for 8.5 h to afford 98 mg (76%) of **11bc** as a liquid.

IR (neat): ν = 2250, 1660 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 3.45 (s, 2 H), 1.88 (s, 3 H), 1.83 (s, 3 H).

MS *m/z* (%): 131 [16.80, $M^+(^{37}Cl)$], 129 [52.53, $M^+(^{35}Cl)$], 67(100).

HRMS: calcd for $C_6H_8{}^{35}ClN$: 129.0345. Found: 129.0339.

Starting from **10b** (44 mg, 0.47 mmol) and $LiCl{\cdot}H_2O$ (43 mg, 0.71 mmol) in HOAc (1.0 mL) at 80 °C, the reaction afforded 11 mg (18%) of **11bc** after 21 h.

3-Iodo-3-octenenitrile (11ca)
Starting from **10c** (120 mg, 0.99 mmol) and NaI (225 mg, 1.5 mmol) in HOAc (0.5 mL) and CF_3CO_2H (0.5 mL), the reaction was carried out at 70 °C for 3.5 h to afford 40 mg (16%) of **(*Z*)-11ca** and 155 mg (63 %) of **(*E*)-11ca**.

(*E*)-**11ca**: liquid.

IR (neat): ν = 2260, 1635 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.37 (t, *J* = 7.49 Hz, 1 H), 3.58 (s, 2 H), 2.02 (q, *J* = 7.23 Hz, 2 H), 1.39–1.18 (m, 4 H), 0.84 (t, *J* = 7.00 Hz, 3 H).

MS *m/z* (%): 249 (60.79, M^+), 56 (100).

HRMS: calcd for $C_8H_{12}IN$: 249.0013. Found: 249.0037.

(*Z*)-**11ca**: liquid.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.94 (t, *J* = 6.80 Hz, 1 H), 3.63 (s, 2 H), 2.15 (q, *J* = 6.87 Hz, 2 H), 1.46–1.26 (m, 4 H), 0.91 (t, *J* = 7.03 Hz, 3 H).

MS *m/z* (%): 249 (89.26, M^+), 81 (100).

Starting from **10c** (89 mg, 0.74 mmol) and NaI (165 mg, 1.1 mmol) in HOAc (1.0 mL) at 70 °C, the reaction afforded 129 mg (70%) of **11ca** (*Z*/*E* = 1:1) after 21 h.

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Starting from **10c** (80 mg, 0.66 mmol) and NaI (149 mg, 0.99 mmol) in CF_3CO_2H (1.0 mL) at 70 °C, the reaction afforded 95 mg (58%) of **11ca** (*Z*/*E* = 1:3.1) after 10 min.

3-Bromo-3-octenenitrile (11cb)
Starting from **10c** (121 mg, 1.0 mmol) and LiBr·H_2O (158 mg, 1.5 mmol) in HOAc (0.5 mL) and CF_3CO_2H (0.5 mL), the reaction was carried out at 70 °C for 6 h to afford 44 mg (22%) of **(***Z***)-11cb** and 118 mg (58%) of **(***E***)-11cb**.

(*E*)-**11cb**: liquid.

IR (neat): ν = 2260, 1650 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.11 (t, *J* = 7.85 Hz, 1 H), 3.55 (s, 2 H), 2.07 (q, *J* = 7.26 Hz, 2 H), 1.45–1.28 (m, 4 H), 0.90 (t, *J* = 7.00 Hz, 3 H).

MS *m/z* (%): 204 [9.46, $M^+ + 1(^{81}Br)$], 202 [10.00, $M^+ + 1(^{79}Br)$], 56 (100).

HRMS: calcd for $C_8H_{12}{}^{81}BrN$: 203.0133. Found: 203.0103.

(*Z*)-**11cb**: liquid.

^1H NMR (300 MHz, $CDCl_3$): δ = 6.08 (t, *J* = 6.37 Hz, 1 H), 3.52 (s, 2 H), 2.20 (q, *J* = 6.71 Hz, 2 H), 1.45–1.24 (m, 4 H), 0.91 (t, *J* = 6.79 Hz, 3 H).

MS *m/z* (%): 204 [22.88, $M^+ + 1(^{81}Br)$], 202 [24.28, $M^+ + 1(^{79}Br)$], 56(100).

3-Chloro-3-octenenitrile (11cc)
Starting from **10c** (121 mg, 1.0 mmol) and LiCl·H_2O (91 mg, 1.5 mmol) in HOAc (0.5 mL) and CF_3CO_2H (0.5 mL), the reaction was carried out at 70 °C for 8 h to afford 35 mg (22%) of **(***Z***)-11cc** and 78 mg (50%) of **(***E***)-11cc**.

(*Z*)-**11cc**: liquid.

IR (neat): ν = 2250, 1665 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.85 (t, *J* = 7.00 Hz, 1 H), 3.38 (s, 2 H), 2.21 (q, *J* = 7.11 Hz, 2 H), 1.41–1.24 (m, 4 H), 0.90 (t, *J* = 6.95 Hz, 3 H).

MS *m/z* (%): 160 [27.96, $M^+ + 1(^{37}Cl)$], 158 [80.24, $M^+ + 1(^{35}Cl)$], 56 (100).

HRMS: calcd for $C_8H_{12}{}^{35}ClN$: 157.0652. Found: 157.0647.

(*E*)-**11cc**: liquid.

IR (neat): ν = 2250, 1660 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.88 (t, *J* = 7.79 Hz, 1 H), 3.41 (s, 2 H), 2.07 (q, *J* = 7.16 Hz, 2 H), 1.40–1.23 (m, 4 H), 0.89 (t, *J* = 6.86 Hz, 3 H).

MS *m/z* (%): 160 [10.81, $M^+ + 1(^{37}Cl)$], 158 [32.61, $M^+ + 1(^{35}Cl)$], 56 (100).

HRMS: calcd for $C_8H_{12}{}^{35}ClN$: 157.0652. Found: 157.0645.

(*E*)-3-Iodo-4-phenyl-3-butenitrile [(*E*)-11da]
Starting from **10d** (38 mg, 0.27 mmol) and NaI (62 mg, 0.41 mmol) in HOAc (0.5 mL) and CF_3CO_2H (0.5 mL), the reaction was carried out at 70 °C for 5 h to afford 32 mg (4%) of (*E*)-**11da**.

(*E*)-**11da**: liquid.

IR (neat): ν = 2240, 1620 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.53 (s, 1 H), 7.43–7.18 (m, 5 H), 3.80 (s, 2 H).

MS *m/z* (%): 269 (73.40, M^+), 115 (100).

HRMS: calcd for $C_{10}H_8IN$: 268.9700. Found: 268.9688.

(*E*)-3-Bromo-4-phenyl-3-butenenitrile [(*E*)-11db]
Starting from **10d** (56 mg, 0.40 mmol) and LiBr·H_2O (62 mg, 0.59 mmol) in HOAc (0.7 mL) and CF_3CO_2H (0.7 mL), the reaction was carried out at 70 °C for 5 h to afford 71 mg (82%) of **(***E***)-11db**.

(*E*)-**11db**: liquid.

IR (neat): ν = 2260, 1635 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.37–7.12 (m, 6 H) 3.64 (s, 2 H).

MS *m/z* (%): 223 [26.90, $M^+(^{81}Br)$], 221 27.65, $M^+(^{79}Br)$], 115 (100).

HRMS: calcd for $C_{10}H_8{}^{79}BrN$: 220.9841. Found: 220.9847.

(*E*)-3-Chloro-4-phenyl-3-butenenitrile [(*E*)-11dc]
Starting from **10d** (64 mg, 0.45 mmol) and LiCl·H_2O (42 mg, 0.69 mmol) in HOAc (0.5 mL) and CF_3CO_2H (0.5 mL), the reaction was carried out at 70 °C for 6 h to afford 30 mg (37%) of (*E*)-**11dc**.

(*E*)-**11dc**: liquid.

IR (neat): ν = 2250,1640 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.37–7.11 (m, 5 H), 6.92 (s, 1 H), 3.50 (s, 2 H).

MS *m/z* (%): 177 [21.03, $M^+(^{35}Cl)$], 179 [9.06, $M^+(^{37}Cl)$], 115(100).

HRMS: calcd for $C_{10}H_8{}^{35}ClN$: 177.0345. Found: 177.0399.

Pd(0)-catalyzed Coupling Reaction of 3-Iodo-4-methyl-3-pentenenitrile (11ba)

4-Methyl-3-phenyl-3-pentenenitrile (7h)
To a solution of PhI (211 mg, 1.03mmol) in THF (1 mL) was added *n*-BuLi (2.0 M, 0.52 mL) at –78 °C while stirring. After the addition, the reaction mixture was warmed up to –40 °C, and a solution of dried $ZnBr_2$ (582 mg, 2.59 mmol) in THF (2 mL) was added. To this solution was added a solution of 3-iodo-4-methyl-3-pentenenitrile **(11ba)** (112 mg, 0.51 mmol) and $Pd(PPh_3)_4$ (30 mg, 0.025 mmol) in THF (1 mL). After stirring at r.t. for 7 h, the solvent was removed by evaporation. Chromatography on silica gel afforded 64 mg (74%) of **7h**.

7h: liquid.

IR (neat): ν = 2248, 1601 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.40–7.17 (m, 5 H), 3.37 (s, 2 H), 1.93 (s, 3 H), 1.65 (s, 3 H).

MS *m/z* (%): 171 (78.04, M^+), 129 (100).

HRMS: calcd for $C_{12}H_{13}N$: 171.10480. Found: 171.10148.

4-Methyl-3-(2-phenyl-1-ethynyl)-3-pentenenitrile (7i)
To a Schlenk reaction tube were added 3-iodo-4-methyl-3-pentenenitrile **(11ba)** (98 mg, 0.44 mmol), MeCN (1 mL), $PdCl_2(PPh_3)_2$ (18 mg, 0.022 mmol), CuI (8 mg, 0.044 mmol), K_2CO_3 (122 mg, 0.89 mmol), phenylacetylene (59 mg, 0.58 mmol), and MeCN (1 mL). This reaction mixture was stirred at r.t. for 20 h, then the solvent was removed by evaporation. Chromatography on silica gel afforded 79 mg (91 %) of **7i**.

7i: liquid.

IR (neat): ν = 2251, 1597 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.41–7.19 (m, 5 H), 3.24 (s, 2 H), 2.02 (s, 3 H), 1.82 (s, 3 H).

MS *m/z* (%): 195 (100, M^+).

HRMS: calcd for $C_{14}H_{13}N$: 195.10480. Found: 195.10282.

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Hydrohalogenation of 1,2-Allenyl Diphenyl Phosphine Oxide

2-Iodo-2-propenyl Diphenyl Phosphine Oxide (13aa)

Starting from **12a** (240 mg, 1 mmol) and LiI (147 mg, 1.1 mmol) in HOAc (1.0 mL), the reaction was carried out at 100 °C for 12 h to afford 276 mg (75%) of **13aa**.

White solid; mp 99–100 °C (CH_2Cl_2–petroleum ether).

IR (KBr): ν = 1600, 1432, 1170, 890 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.85–7.68 (m, 4 H), 7.62–7.36 (m, 6 H), 6.42–6.34 (m, 1 H), 5.98–5.90 (m, 1H), 3.62 (d, $^2J_{H,P}$ = 13.0 Hz, 2 H).

MS *m/z* (%): 569 (1.84, M^+ + 201), 368 (2.94, M^+), 367 (2.63, M^+ – 1), 201 (100).

Anal. Calcd for $C_{15}H_{14}IOP$: C, 48.94; H, 3.83. Found: C, 48.92; H, 3.75.

Starting from **12a** (240 mg, 1.0 mmol), NaI (165 mg, 1.1mmol) and HOAc (1.0 mL), the mixture was heated under reflux for 23 h to afford 220 mg (60%) of **13aa**.

Starting from **12a** (120 mg, 0.5 mmol) and hydriodic acid (45%, 1 mL), the mixture was heated under reflux for 48 h to afford 85 mg (4 %) of **13aa**. As eluent for chromatographic separation petroleum ether–ethyl acetate (2:1) was used.

2-Bromo-2-propenyl diphenyl phosphine oxide (13ab)

Starting from **12a** (240 mg, 1.0 mmol) and $LiBr·H_2O$ (114 mg, 1.1 mmol), the reaction was carried out at 100 °C for 36 h to afford 225 mg (70%) of **13ab**.

White solid; mp 139–140 °C (CH_2Cl_2–petroleum ether).

IR (KBr): ν = 1610, 1430, 1170, 890 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.85–7.68 (m, 4 H), 7.62–7.36 (m, 6 H), 5.95–5.85 (m, 1 H), 5.65–5.55 (m, 1 H), 3.56 (d, $^2J_{H,P}$ = 13.1 Hz, 2 H).

MS *m/z* (%): 523 [12.96, M^++201(^{81}Br)], 521 [12.04, M^++201(^{79}Br)], 322 [7.91, M^+(^{81}Br)], 320 [8.14, M^+(^{79}Br)].

Anal. Calcd for $C_{15}H_{14}BrOP$: C, 56.10; H, 4.39. Found: C, 56.36; H, 4.36.

Starting from **12a** (120 mg 0.5 mmol) and 40 % HBr (1.0 mL), the mixture was heated under reflux for 60 h to afford 60 mg (37%) of **13ab**.

2-Chloro-2-propenyl Diphenyl Phosphine Oxide (13ac)

Starting from **12a** (240 mg, 1.0 mmol) and $LiCl·H_2O$ (67 mg, 1.1 mmol), the reaction was carried out at 100 °C for 40 h to afford 160 mg (58%) of **13ac**.

White solid; mp 149–150 °C (CH_2Cl_2–petroleum ether).

IR (KBr): ν = 1610, 1430, 1170, 890 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.85–7.68 (m, 4 H), 7.62–7.36 (m, 6 H), 5.45–5.35 (m, 1 H), 5.35–5.28 (m, 1 H), 3.44 (d, $^2J_{H,P}$ = 13.5 Hz, 2 H).

MS *m/z* (%): 479 [12.71, M^+ + 201(^{37}Cl)], 477 [35.08, M^+ + 201(^{35}Cl)], 278 [5.59, M^+(^{37}Cl)], 276 [19.12, M^+(^{35}Cl)], 201 (100).

Anal. Calcd for $C_{15}H_{14}ClOP$: C, 65.11; H, 5.10. Found: C, 65.30; H, 5.04.

Starting from **12a** (120 mg 0.5 mmol) and aqueous HCl (36 %, 1.0 mL), the mixture was heated under reflux for 60 h to afford 43 mg (37%) of **13ac**.

Pd(0)-catalyzed Coupling Reaction of 2-Iodo-2-propenyl Diphenyl Phosphine Oxide (13aa)

2-Methylene-3-octynyl Diphenyl Phosphine Oxide (7j)

A mixture of **13aa** (184 mg, 0.5 mmol), *n*-hexyne (49 mg, 0.6 mmol), $PdCl_2(PPh_3)_2$ (3 mg, 0.005 mmol), CuI (1.0 mg, 0.005 mmol) and Et_3N (101 mg, 1.0 mmol) in 2 mL CH_3CN was stirred at r.t. overnight. After complete conversion of the starting material as monitored by TLC, the mixture was quenched with H_2O and extracted with CH_2Cl_2. The combined CH_2Cl_2 layers were dried over $MgSO_4$. Evaporation of the solvent and flash chromatography on silica gel (eluent: petroleum ether–ethyl acetate, 2:1) afforded 150 mg (93%) of **7j**.

White solid; mp 69–70 °C (CH_2Cl_2–petroleum ether).

IR (KBr): ν = 2235, 1606, 1435, 1191 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.85–7.68 (m, 5 H), 7.62–7.36 (m, 5 H), 5.36 (d, $^2J_{H,H}$ = 4.2 Hz, 1 H), 5.31 (d, $^2J_{H,H}$ = 4.2 Hz, 1 H), 3.18 (d, $^2J_{H,P}$ = 14.0 Hz, 2 H), 1.96 (t, $^2J_{H,H}$ = 6.8 Hz, 2 H), 1.30–1.10 (m, 4 H), 0.82 (t, $^2J_{H,H}$ = 6.8 Hz, 3 H).

MS *m/z* (%): 322 (4.88, M^+), 321 (5.66, M^+-1), 280 (100).

Anal. Calcd for $C_{21}H_{23}OP$: C, 78.24; H, 7.19. Found: C, 78.11; H, 7.19.

2-Methylene-4-phenyl-3-butynyl diphenyl phosphine oxide (7k)

Starting from **13aa** (184 mg, 0.5 mmol), phenylacetylene (63mg, 0.6 mmol) $PdCl_2(PPh_3)_2$ (3 mg, 0.005 mmol), CuI (1.0 mg, 0.005 mmol) and Et_3N (101 mg, 1.0 mmol) in 2 mL CH_3CN, the reaction was carried out at r.t. overnight to afford 162 mg (95%) of **7k**.

Light-yellow solid; mp 115–116 °C (EtOAc–petroleum ether).

IR (KBr): ν = 1994, 1603, 1440, 1185 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.85–7.68 (m, 5 H), 7.62–7.36 (m, 5 H), 7.30–7.15 (m, 3H), 7.15–7.05 (m, 2 H), 5.56 (d, $^2J_{H,H}$ = 4.2 Hz, 1 H), 5.51 (d, $^2J_{H,H}$ = 4.2 Hz, 1 H), 3.30 (d, $^2J_{H,P}$ = 13.9 Hz, 2 H).

MS *m/z* (%): 342 (52.74, M^+), 341 (78.65, M^+-1), 201 (100).

Anal. Calcd for $C_{23}H_{19}OP$: C, 80.69; H, 5.59. Found: C, 80.23; H, 5.65.

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